Palestine Technical University – Kadoorie
Faculty of Engineering and Technology
Department of Computer Systems Engineering

Project title:

DESIGNING A GRADUATION PROJECTS MANAGEMENT SYSTEM FOR FACULTY OF ENGINEERING AND TECHNOLODY AT PALESTINE TECHNICAL UNIVERSITY-KADOORIE

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الإهداء

بسم الله الرحمن الرحيم

" وَقِيلَ الْحَمْدُ لِلَّهِ رَبِّ الْعَالَمِينَ "

الحمد لله الذي علم بالقلم علم الإنسان ما لم يعلم والصلاة والسلام على خير البرية ومعلم البشرية سيدنا محمد وعلى آله وصحبه وسلم، وبعد....

إلى الأم الرؤوم حاضنة الأبطال وموطن الشامخين، إلى الصامدين رغم أنف الطغاة أطماع الحاقدين

غزة العزة بأبطالها ومجاهديها وشهدائها وأسراها

إليها بكلّ من فيها وما فيها.

إلى العبق الزكي والأرواح الطاهرة ،إلى من خضبوا الأرض بدمائهم لتحيا أبية ونحيا أعزة،

شهداء غزة وجنين طولكرم ونابلس وكلّ فلسطين

إلى الأسود القابعة في زنازين الظلم، أسرانا البواسل، فك الله بالعز قيدكم.

إلى من نفضوا عنا غبار العثرات وضمدوا الجراحات وتجاوزوا عن الزلّات وشيّدوا قلاع دعمٍ لا متناهٍ وغرسوا في عقولنا أسمى القيم، آبائنا وأمهاتنا.

إلى مؤنسي الرحلة وداعميها بكلّ محطاتها، إلى الأحنّ علينا منّا

إخوتنا وأخواتنا.

إلى من نثروا شذى علمهم في مسيرتنا، وجادوا فما بخلوا ولا كلّوا

العقول النيّرة دكاترتنا وأستاذتنا وكل من فاض نهر عطائه على ضفاف طريقنا.

إلى رفاق الدرب ومؤازريه، وخير ما أهدتنا الأيام إلى أصدقائنا وزملائنا وكلّ من أبصرنا أثر خطواته في مسيرنا .

نهدي ثمرة جهدنا ...

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In the end, we would like to thank the group members for their respect, cooperation, helping each other with kindness, and completing their work efficiently.

ABSTRACT

Graduation year is one of the important years in a student's academic life, so it is necessary to help him pass it with the best results.

When we reached the beginning of the graduation year, we faced several difficulties, including finding a supervisor for our project. We discovered that all the students had reserved their supervisor at the end of the previous semester, and thus most of the supervisors became unavailable.

Moreover, we found it difficult to research whether this idea had been applied previously due to the lack of information about the projects that were completed at our university in previous years.

After we chose a supervisor for the project, we noticed difficulty communicating with the supervisor and distraction between the LMS and social media.

Our site will include an admin who will be responsible for adding sections and adding Students, supervisors, and department heads, review their data and modify or delete it.

In addition, he will add, review or delete previous projects.

The visitor to our website will be able to review the previous projects for each section and ask the admin about any question

The department head will be responsible for adding sections to his department, in addition to reviewing or deleting them.
Through our website, the student will be able to book a section and wait for supervisor approval of the request.

The supervisor will be able to review requests and approve or reject them. After approval, the student will be able to communicate with his supervisor and members of his group, in addition to solving the tasks that the supervisor will assign to each section separately and obtaining feedback from the supervisor.

الملخص

تعتبر سنة التخرج من الأعوام المهمة في الحياة الأكاديمية للطالب، لذا لا بد من مساعدته على اجتيازها بأفضل النتائج.

وعندما وصلنا إلى بداية عام التخرج، واجهنا عدة صعوبات، منها إيجاد مشرف لمشروعنا. اكتشفنا أن جميع الطلاب قد حجزوا مشرفهم في نهاية الفصل الدراسي السابق، وبالتالي أصبح معظم المشرفين غير متوفرين.

علاوة على ذلك، وجدنا صعوبة في البحث فيما إذا كانت هذه الفكرة قد تم تطبيقها سابقاً نظراً لقلة المعلومات حول المشاريع التي تم إنجازها في جامعتنا في السنوات السابقة.

بعد أن اخترنا مشرفاً للمشروع، لاحظنا صعوبة في التواصل مع المشرف وتشتت الانتباه بين نظام إدارة التعلم (LMS) ووسائل التواصل الاجتماعي.

سيضم موقعنا آدمن سيكون مسؤولاً عن إضافة الأقسام وإضافة الطلاب والمشرفين ورؤساء الأقسام ومراجعة بياناتهم وتعديلها أو حذفها. بالإضافة إلى ذلك، سيقوم بإضافة أو مراجعة أو حذف المشاريع السابقة.

سيتمكن زائر موقعنا من مراجعة المشاريع السابقة لكل قسم وسؤال الآدمن عن أي سؤال. وسيكون رئيس القسم مسؤولاً عن إضافة الأقسام إلى قسمه بالإضافة إلى مراجعتها أو حذفها.

ومن خلال موقعنا سيتمكن الطالب من حجز الشعبة وانتظار موافقة المشرف على الطلب.

سيتمكن المشرف من مراجعة الطلبات والموافقة عليها أو رفضها. بعد الموافقة سيتمكن الطالب من التواصل مع مشرفه وأعضاء مجموعته، بالإضافة إلى حل المهام التي سيكلفها المشرف لكل شعبة على حدة والحصول على التغذية الراجعة من المشرف.

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CHAPTER ONE

INTRODUCTION

1.1 Overview

In our current era, the Internet and websites are relied upon greatly to accomplish all life's requirements in an easy and safe manner, and without any significant effort. from here came the idea of the website for computing the web system for managing graduation projects.

At the end of the graduation year, students work on their projects, and most of these projects are not developed even though some of these projects are huge and deserve to be developed. Students do not know whether a particular idea has been implemented or not, in addition to the burden placed on the teaching staff, such as following up with the different groups for each supervisor, and finding ways to organize the project process and communicate with students, especially in difficult circumstances such as the ones we are currently experiencing.

The Graduation Projects Management System is a useful tool for managing academic projects. It offers secure logins and controlled access for students, faculty, and administrators. Key features include creating projects, tracking tasks, collaborating, and managing documents, making project coordination easy and efficient.

1.2 Problem Statement

Students face challenges in choosing a supervisor for their graduation project. In addition to the difficulty of accessing previous students' projects, and knowing what has been implemented and to what stage it has been reached. The lack of a unified means of communication with the supervisor, and the absence of a specific place to submit parts of thesis that the supervisor requests to obtain continuous evaluation.

1.3 General Objective

Our general goal with the Graduation Projects Management System is to create an effective and easy-to-use platform that improves the management of academic projects for department heads, supervisors, and students.

1.4 Objectives

- Building a website that helps students choose a supervisor for their graduation project.
- The supervisor can set assignments for the student to submit.
- It contains a chatbot to answer common student inquiries.
- Facilitating the process of managing sections by the department head in terms of adding and deleting them.
- Provide continuous and effective feedback from professors, contributing to improving the quality of the projects and supporting students in achieving their academic goals.
- Enable direct communication between students supervisor and his/her groups through an effective messaging system, allowing students to ask questions and share their progress.
- Provide a user-friendly interface enabling students to review previous students' projects, and access their theses.

1.5 Scope of Work

This website will help students of the Faculty of Engineering at our university who are about to graduate, supervisors and department heads, and it can also be developed to include all colleges of the university.

1.6 Process Model

We know that the project will go through a lot of changes in both functionality and design throughout the development time, and these changes happen due to three reasons, stated below:

- We have a basic idea of what the project should look like and do, but we haven't figured out all the little details yet, which means there may be design changes in the future.
- Because we'll be getting familiar with new programming frameworks for various project parts like front-end and back-end, as time goes on, we'll get better at using these frameworks. This will allow us to come up with improved and more efficient solutions and find different approaches to solving problems.
- We want to be able to introduce a new feature or remove one if we think it's required.

So we decided to choose a software model that is flexible and responds quickly to changes compared to other models, which is the agile model due to its features :

- Customer satisfaction by rapid, continuous delivery of useful software.
- People and interactions are emphasized rather than processes and tools. Customers, developers and testers constantly interact with each other.
- Working software is delivered frequently (weeks rather than months).
- Face-to-face conversation is the best form of communication.
- Close, daily cooperation between businesspeople and developers.
- Continuous attention to technical excellence and good design.
- Regular adaptation to changing circumstances.

- Even late changes in requirements are welcomed.

Agile provides different models each one provides different techniques and tools to apply and achieve agile model principles, but in our case, we chose the Scrum model due to its compatibility with Agile principles, offering varied techniques and tools for effective implementation and adherence.

-Its user-friendly nature is due to the simplicity of providing general instructions, allowing for custom implementation – an easier alternative to mastering a new workflow.

-Moreover, it exhibits quick adaptability to changes, effortlessly incorporating new adjustments into the backlog for future consideration in planning.

Our project implementation revolves around the Agile process model, characterized by iterative development in software development. The plan is to segment project tasks into brief two-week iterations, referred to as scrums. This approach of dividing the entire project into scrums is designed to minimize project risks and streamline the overall project delivery timeline.[1.1]

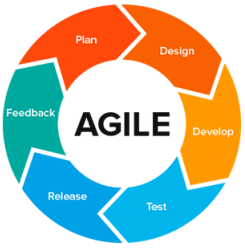

Figure 1 Agile Method Life Cycle

CHAPTER TWO

LETRETURE REVIEW

2.1 The Graduation Project Management (Palestine Technical University – Kadoorie 2021)

System (GPMS) will certainly perform these educational tasks, as the site will serve the student and the supervisor.

for the students, the site will help to choose the team, idea and supervisor, and also facilitate the communication process between the supervisor and students and between students and each other, and also provides a feature to deliver reports to the supervisor, GPMS is equipped with a data analysis tool to monitor student achievement.

2.2 Graduation project Content Management System (Princess Nourah Bent Abdurrahman University 2015)

The project provides the proposed Content Management graduation project since the first phase of work to make the final document of the project, so it is considered the main reference for students, which contains all the stages of the work required, and the models and how to implement the project and all that is required of students in this area.[2.1]

2.3 Graduation-project management system (University of Petra, Jordan 2017)

The main objective of this system is to manage and control activities such as submission of project deliverables, grading, evaluation, announcements, and supplementary course material along with facilitate an efficient communication between the many participants in the graduation project course including: students, supervisors, mentors, examiners and graduation project committee members. While students canorganize their deadline submissions with lesseffort, supervisors and graduation project committee can track and monitor students transactions instantaneously.[2.2]

CHAPTER THREE

SYSTEM REQUIREMENTS

3.1 Functional Requirements

Functional requirements define what a product must do and what its features and functions are.[3.1]

3.1.1 ADMIN

- The admin must be able to login to the website
- The admin must be able to add,view ,edit or delete students,supervisors, department heads.
- The admin must be able to add or delete departments.
- The admin must be able to add or delete previous projects
- The admin must be able to send announcements to the students.
- The admin must be able to review some statistical information regarding the number of previous projects in each department, in addition to the number of students, supervisors, and department heads within the college.
- The system must provide the admin with the ability to access and see previous projects.

3.1.2 Head of Department

- Head of department must be able to login the website.
- The head of department must be able to view the personal page and modify it as appropriate, with the possibility of changing the password, image and phone number.
- The system must allow the head of department to add, delete and view sections.
- The system must provide the head of departmet with the ability to access and see previous projects.

3.1.3 Student

- Student must be able to login the website.
- The system must give the student the ability to choose the appropriate supervisor
- The student must be able to view the personal page and modify it as appropriate, with the possibility of changing the password, image and phone number.
- The system must provide the student with the ability to access and see previous projects.
- The system must allow the student to communicate with the supervisor through his own chat.
- the student must be able to review supervisor details and his group information.
- The system must allow the student to submit assignments set by the supervisor, edit it and get feedback.
- The system must provide a chatbot to answer frequently asked student questions about writing thesis.

3.1.4 Supervisor

- Supervisor must be able to login the website.
- The supervisor must be able to view the personal page and modify it as appropriate, with the possibility of changing the password , image and phone number.
- The system must provide supervisor the ability to see his/her assigned students
- Supervisor must be able to create,assign, and assess project tasks for students
- Supervisor must be able to communicate with students through chat.
- Supervisor must be able to view and accept or reject students requests.

- The system must provide the supervisor with the ability to access and see previous projects.

3.2 Non-Functional Requirements

Non-functional requirements describe the general properties of a system. They are also known as quality attributes.[3.1]

- **Usability:**

- The user interacts with the site through a web page programmed using React js and Material UI.
- User friendly interface, easy to learn and easy to use.

- **Security:**

- Every user has a password-protected account, so no one can access it using their account without the correct password.
- The user's password is encrypted in the database, so if someone gains access to the database, he will not be able to decrypt this data. This is done by using the becrypt library, bcrypt library take the password and encrypt it 4 times; hash password = bcrypt. hash (password, saltround).

- **Portability:**

- Since our website is designed to work as a web page on the browser through the Internet, it is therefore able to work on multiple systems such as Android, Windows, Linux, etc. Therefore, its operation on different systems and changing the work environment will not require any additional costs.

- **Performance:**

- The system should respond to user actions within a reasonable time frame to provide a smooth user experience.

- **Availability:**

- the system should always be available for access at 24 hours, 7 days a week.

- **Maintenance:**

  - The system should be easy to maintain and update with minimal disruption to users.
  - Regular updates and maintenance should be scheduled during low-usage periods.

CHAPTER FOUR

METHODOLOGY AND TECHNOLOGY

4.1 Technology Stack

A technology stack is a set of technologies you choose and use to build a web application, mobile application or similar applications. A good technology stack must give a seamless user experience as well as be scalable and cost-effective. A typical tech stack contains a frontend, backend and database and is known as a full technology stack.[4.1]

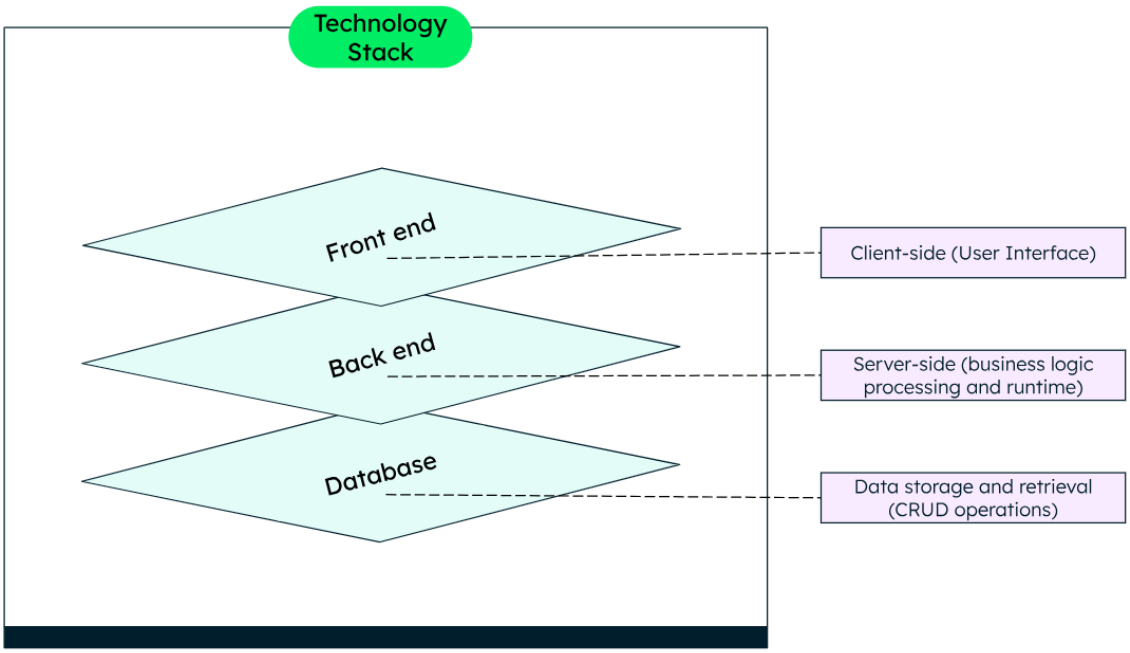

Figure 2: Technology Stack

4.1.1 MERN Stack

MERN is a pre-built technology stack based on JavaScript technologies. MERN stands for **MongoDB, Express, React, and Node**, after the four key technologies that make up the stack.[4.1]

MongoDB — document database

Express(.js) — Node.js web framework

React(.js) — a client-side JavaScript framework

Node(.js) — the premier JavaScript web server (runtime)

4.2 Integrated Development Environment

4.2.1 Visual Studio Code (Vscode)

For the implementation of the Correspondence & Workflow Management System prototype,we used VSCode to write The needed markup and script code for the website. Particularly, Microsoft Visual Studio Code 2022 will be used in the development process.

VSCode already supports a plethora of different programming syntaxes out of the box, but

you still can install a language extension to expand support for useful features, such as IntelliSense auto-complete and advanced linting. The sky is the limit, from Python and C++ to COBOL and Lisp. All you have to do is click on an “Install” button right inside VSCode. [4.2]

Figure 3: VS code logo

4.3 Front-End Technologies

It is an aspect of a web application that is available to a customer and is responsible for the user interface, allowing them to communicate with the application through a convenient and easy-to-use interface. Front-end development involves creating the visual components of the website or application that users see and interact with, such as buttons, forms, menus, and other graphic elements.

4.3.1 React JS

React is an open-source, front end, JavaScript library for building user interfaces or UI components. It is maintained by Facebook, and a community of individual developers and companies. [4.3]

Figure 4: React logo

4.3.1.1 Why React?

React is known to be a versatile tool that can be utilized on both desktop and mobile platforms with many distinct features. One of them is the ability to create interactive rather than simple static pages that can update and render data after each input from the user, thus allowing for a seamless UI without the need to refresh the entire page every time a change is made. The reason for this is due to React's structure is based on components that allow for the design of complex UIs due to the component logic written in JavaScript.

4.3.2 React Material UI (MUI)

React Material UI is a framework built upon React library which contains predefined components to create React applications. Material UI is basically a design language built by Google in 2014 and works with various JavaScript frameworks apart from React such as Angular.js and Vue.js.

Figure 5: React MUI logo

4.3.2.1 Why MUI?

The quality of the inbuilt designs of Material UI and its easy implementation makes it the first choice of most developers. The inbuilt components are also customizable, so it helps in easily recreating the designs.

4.3.3 JavaScript

JavaScript (JS) is a computer programming language used to make websites and applications dynamic and interactive.It's unique because it can run directly in your browser, not just on a server.

Developers use JS to add animations, pop-up windows, search bars, buttons, audio and video, chat widgets, and other interactive elements to a webpage.

You can also use JS to provide real-time updates to a page without reloading it.

For example, websites with news tickers, stock or crypto prices, or real-time product availability information typically use JavaScript to add this functionality.

Figure 6: JavaScript logo

It can be used for client-side and server-side, frontend and back end. [4.4]

4.3.4 CSS

Cascading Style Sheets (CSS) is a language that defines the presentation of a document. CSS can also be used with most markup languages, including XUL and SVG, and with practically any XML document that supports style sheets. CSS has been a fundamental part of the Web stack for the past couple of years, and like other technologies, it presents several security challenges, it controls the layout of a web site for various media. A web page could be resized or modified depending on whether it's being rendered in a browser, a mobile phone, or sent to a printer. [4.5]

Figure 7: CSS logo

4.4 Back-End Technologies

A back-end framework is used to create server-side web architectures stably and efficiently. Backend technologies focus on improving the hidden aspects of the websites and are very important for the proper execution of any web application. It enables the smooth creation and maintenance of a website.

4.4.1 Node.js

Node (or more formally *Node.js*) is an open-source, cross-platform runtime environment that allows developers to create all kinds of server-side tools and applications in JavaScript. [4.6]

Figure 8: Node.js logo

From a web server development perspective Node has a number of benefits:

- Great performance! Node was designed to optimize throughput and scalability in web applications and is a good solution for many common web-development problems (e.g. real-time web applications).
- Code is written in "plain old JavaScript", which means that less time is spent dealing with "context shift" between languages when you're writing both client-side and server-side code.
- JavaScript is a relatively new programming language and benefits from improvements in language design when compared to other traditional web-server languages (e.g. Python, PHP, etc.) Many other new and popular languages compile/convert into JavaScript so you can also use TypeScript.
- The node package manager (npm) provides access to hundreds of thousands of reusable packages. It also has best-in-class dependency resolution and can also be used to automate most of the build toolchain.

- Node.js is portable. It is available on Microsoft Windows, macOS, Linux, etc Furthermore, it is well-supported by many web hosting providers, that often provide specific infrastructure and documentation for hosting Node sites.

4.4.2 Express.js

Express.js is a fast, flexible and minimalist web framework for Node.js. It's effectively a tool that simplifies building web applications and APIs using JavaScript on the server side. Express is an open-source that is developed and maintained by the Node.js foundation.

Express.js offers a robust set of features that enhance your productivity and streamline your web application. It makes it easier to organize your application's functionality with middleware and routing. It adds helpful utilities to Node HTTP objects and facilitates the rendering of dynamic HTTP objects.[4.7]

Figure 9: Express.js logo

4.4.3 MongoDB

MongoDB is an open source NoSQL database management program. NoSQL (Not only SQL) is used as an alternative to traditional relational databases. NoSQL databases are quite useful for working with large sets of distributed data. MongoDB is a tool that can manage document-oriented information, store or retrieve information.

MongoDB is used for high-volume data storage, helping organizations store large amounts of data while still performing rapidly

Instead of using tables and rows as in relational databases, as a NoSQL database, the MongoDB architecture is made up of collections and documents.[4.8]

Figure 10: MongoDB logo

4.4.4 Mongoose

Mongoose is an Object Data Modeling (ODM) library for MongoDB and Node.js. It manages relationships between data, provides schema validation, and is used to translate between objects in code and the representation of those objects in MongoDB. [4.9]

Figure 11: Mongoose logo

4.5 Powerful Tools

4.5.1 Postman

Postman is an API(application programming interface) development tool that helps to build, test and modify APIs. Almost any functionality that could be needed by any developer is encapsulated in this tool. It is used by over 5 million developers every month to make their API development easy and simple. It has the ability to make various types of HTTP requests(GET, POST, PUT, PATCH), save environments for later use, converting the API to code for various languages(like JavaScript, and Python).[4.10]

Figure 12: Postman logo

4.5.2 Git & GitHub

GitHub is a cloud-based platform where you can store, share, and work together with others to write code, Storing your code in a "repository" on GitHub allows you to:

- Showcase or share your work.
- Track and manage changes to your code over time.
- Let others review your code, and make suggestions to improve it.
- Collaborate on a shared project, without worrying that your changes will impact the work of your collaborators before you're ready to integrate them.

Git is a version control system that intelligently tracks changes in files. Git is particularly useful when you and a group of people are all making changes to the same files at the same time. [4.11]

Figure 13: Git & GitHub logo

4.5.3 Botsonic

Botsonic is a versatile AI chatbot builder designed for users without coding expertise. Powered by GPT-4, it offers natural conversation abilities, sentiment recognition, and adaptability, learning from user interactions to provide tailored responses. Its no-code nature allows easy chatbot development and deployment, breaking down technical barriers for small to medium-sized businesses. Bosonic stands out for its high customizability, reflecting a brand's unique identity through conversation flows, names, and avatars. The platform is not just limited to creating chatbots but also enables users to match the chatbot's style with their brand, making the interactions more cohesive and tailored to specific business needs. Additionally, it's trainable on custom data, allowing businesses to input their specific knowledge for more accurate responses. The platform provides chat data analysis for continuous improvement and maintains high-security standards. Building a chatbot with Botsonic involves signing up,

uploading a knowledge base, customizing the bot, and integrating it into a website, with analytics to track usage.

CHAPTER FIVE

SOFTWARE ANALYSIS AND DESIGN

5.1 Use Case Diagrams

5.1.1 Admin

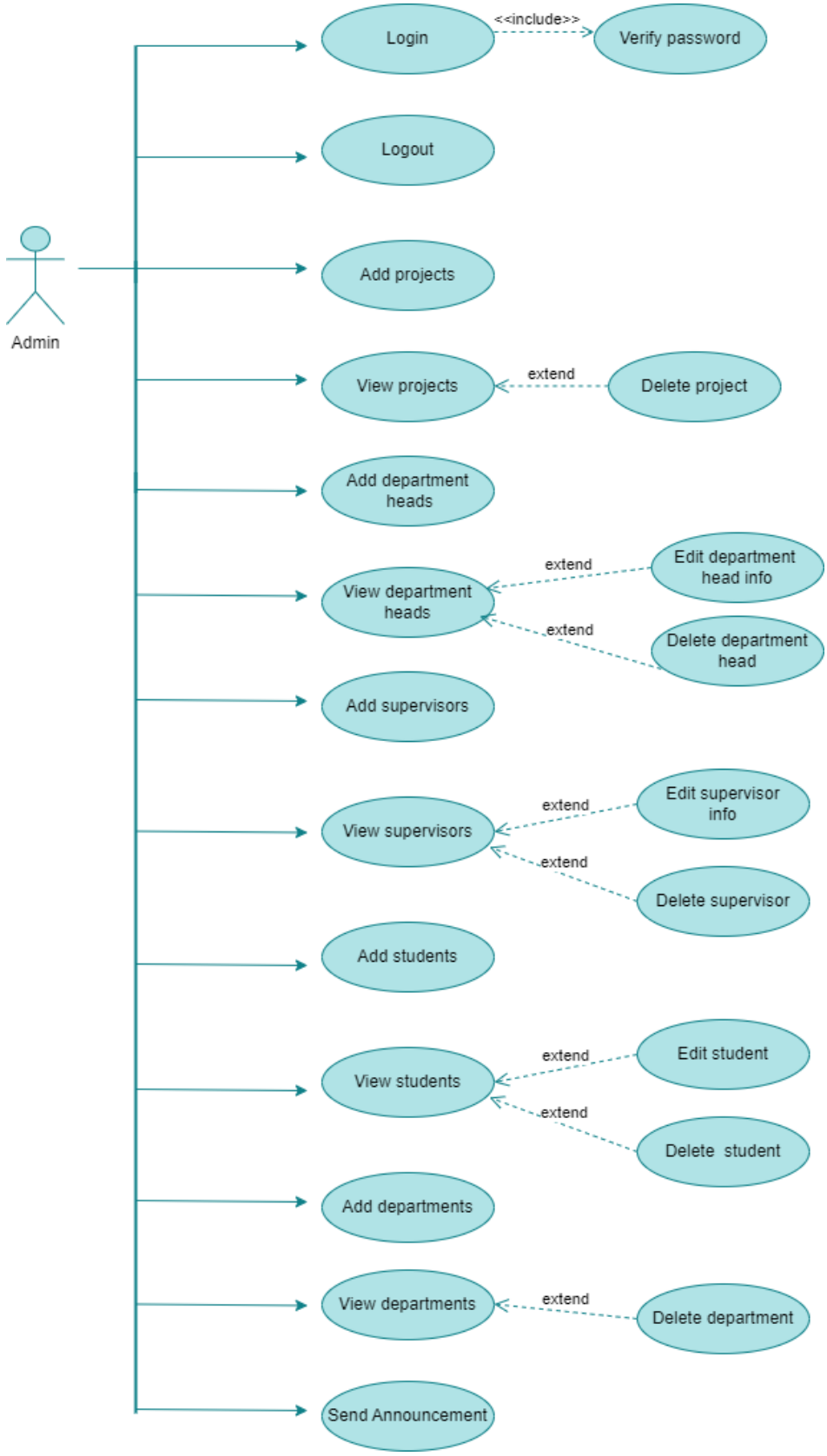

Figure 14: Admin use case diagram

5.1.2 Head of Department

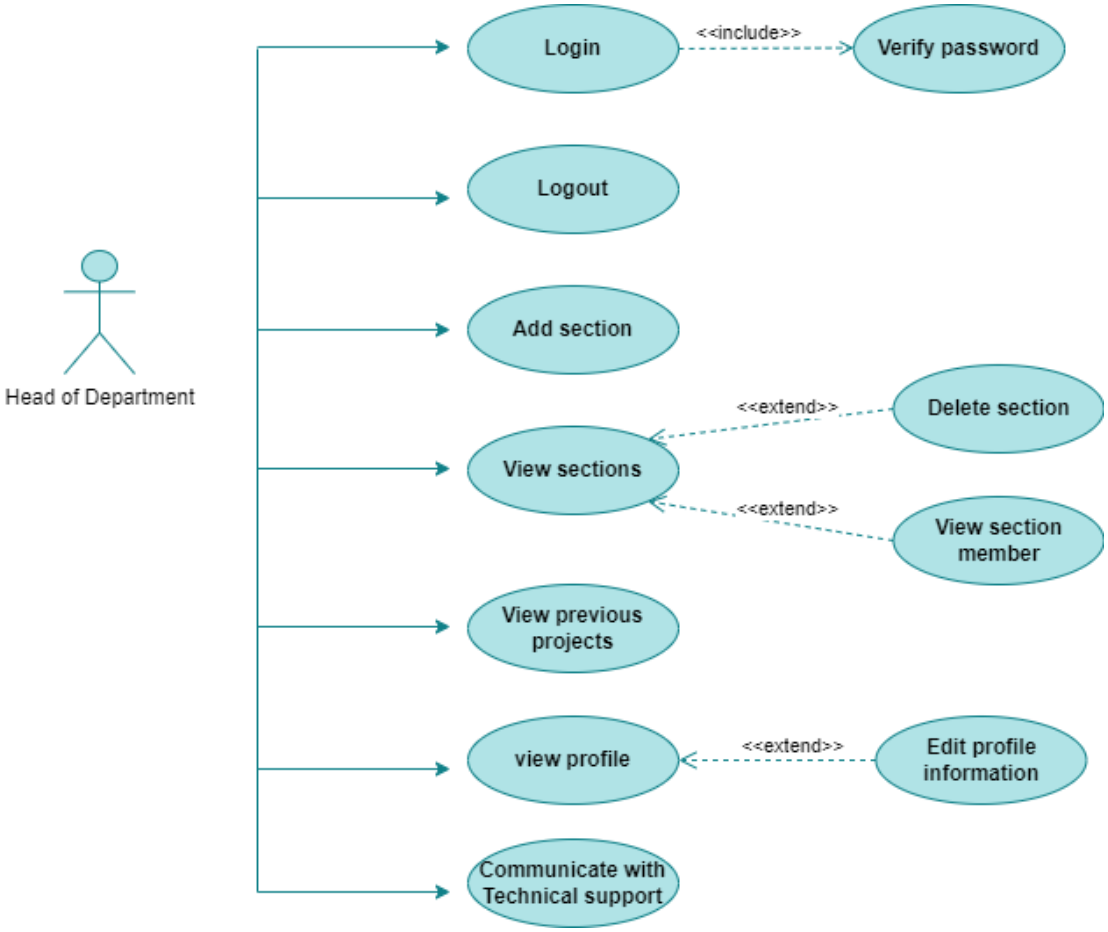

Figure 15: Head of department use case diagram

5.1.3 Supervisor

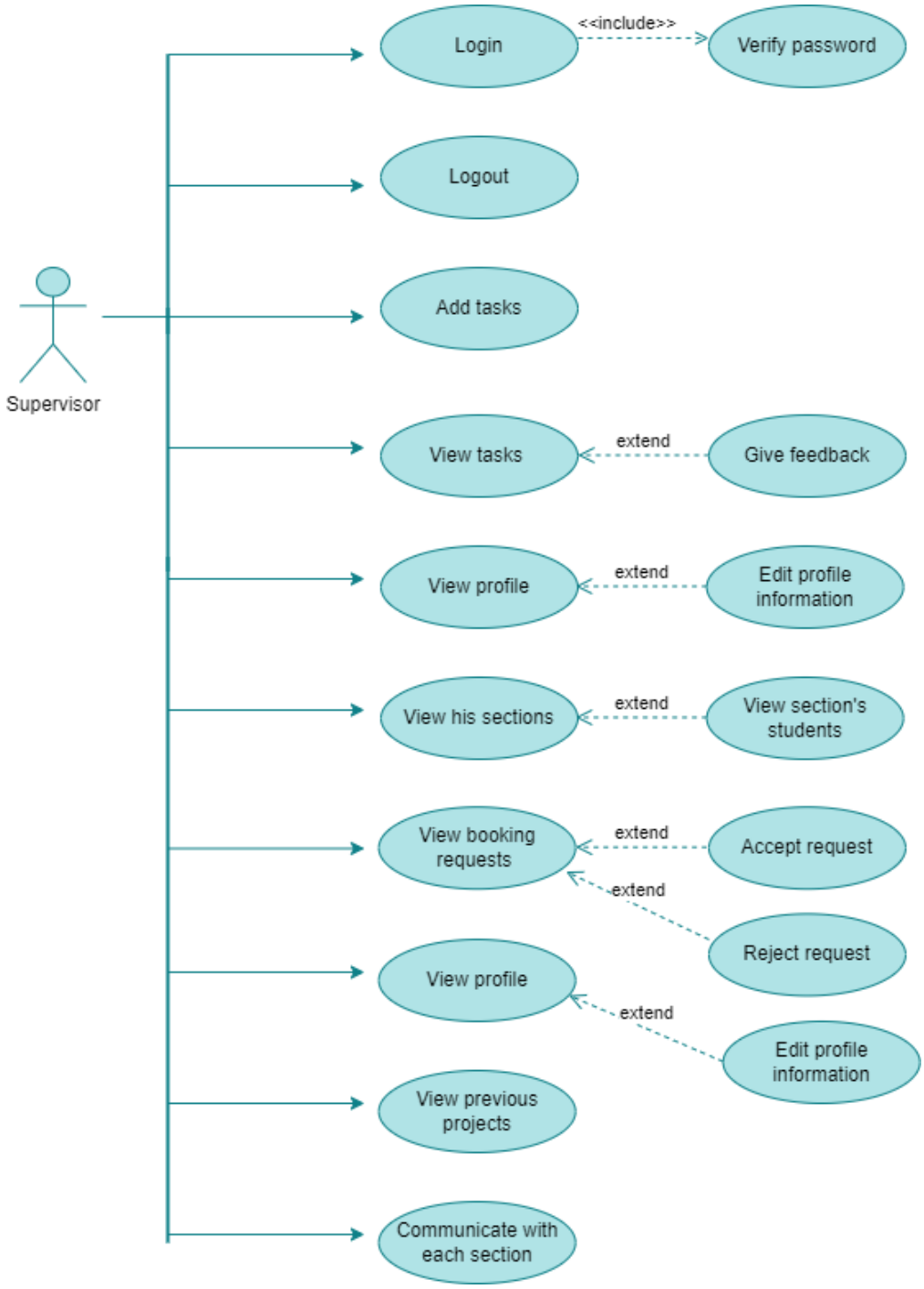

Figure 16: Supervisor use case diagram

5.1.4 Student

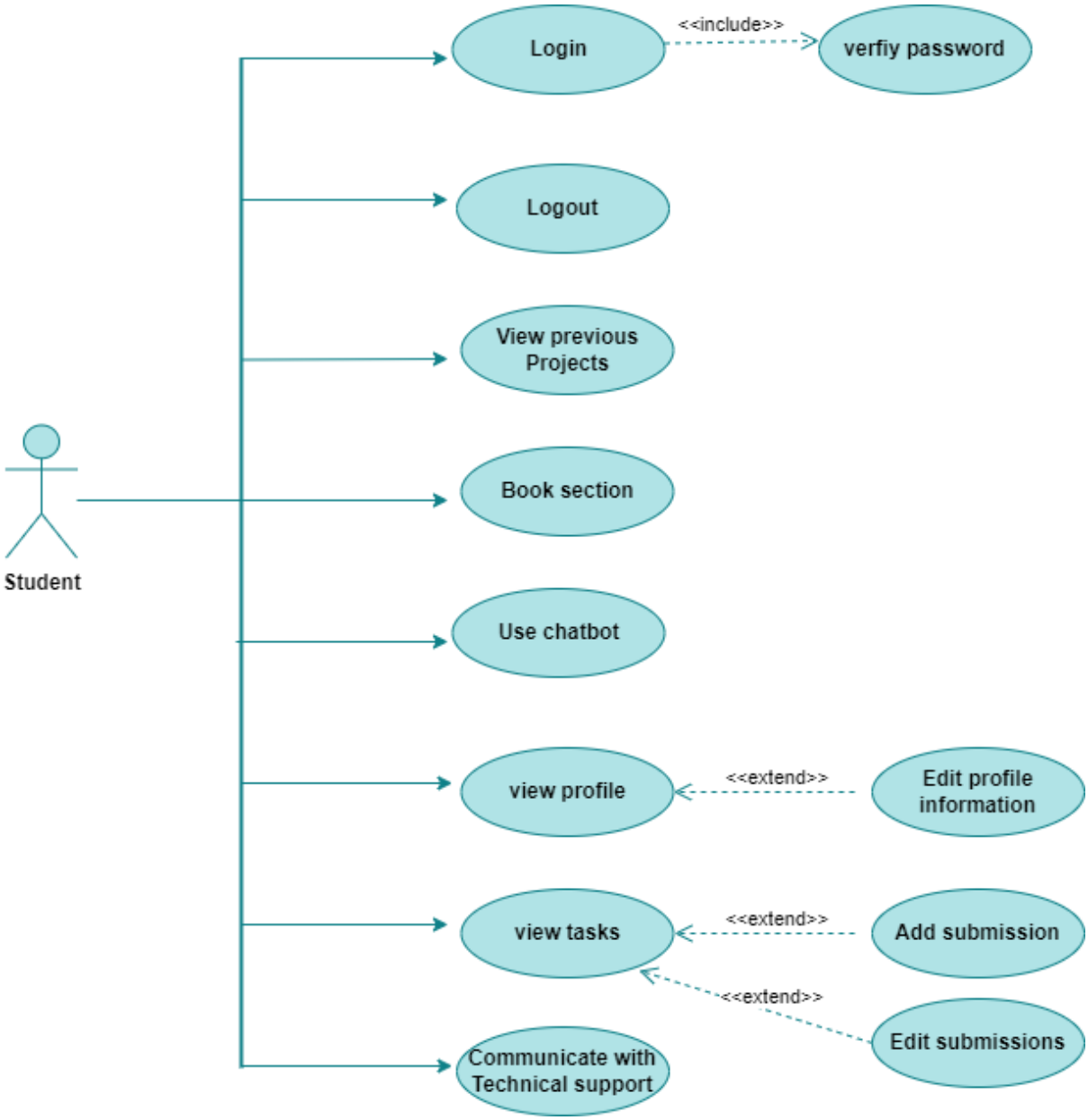

Figure 17: Student use case diagram

5.2 Fully Developed Use Case Description

5.2.1 Register new student

Use case name	Register new student
Scenario:	Create account for a student inside faculty of engineering and technology.
Triggering event:	A student registers the introduction to graduation project course.
Brief description:	Admin creates account for the new student using and entering basic information about the student.
Actors	Admin, student
preconditions	student information must be existed. student authorization must be available
postconditions	1. account must be created. 2. Student's information must be validated. 3. student must be saved and recognized.
Flow of activities	<table><tr><th>Actor</th><th>System</th></tr><tr><td>1. Admin intents to create an account for the new student. 2. Admin enters the needed student's information.</td><td>1.1 system asks for the student's basic information. 2.1 system creates the new account for the student. 2.2 system verifies authorization for student. 2.3 system return valid Student account.</td></tr></table>
Exception conditions:	1. Student's information is incomplete. 2. Student's email is already taken.

Figure 18: Register user fully developed use case description

5.2.2 Update supervisors' information

Use case name	Update supervisor's information
Scenario	Update some old information into new information
Triggering event:	An admin wants to change an information to a specific supervisor
Brief description:	This use case describes how an admin can update supervisors's information: Information: name, email, phoneNumber, officeHours, department, image, password
Actors	Admin
preconditions	Supervisor account subsystem must be available.
postconditions	The new entered data must be validated in the server-side New Information must be saved.
Flow of activities	

Actor	System
1. Admin clicks on the "Supervisor" button link which is found on the sidebar.	1.1 system displays table which is contain all supervisors.
2. Admin chooses the supervisor he is looking for by clicking on edit icon	2.1 system navigates admin to update page which is contain previous information.
3. Admin fills out all required fields in the form and update the information he wants to update.	4.1 system save updated information
4. Admin hits "submit" button	

Exception conditions:	1. The new entered data must be validated in the server-side 2. New Information must be saved.

Figure 19: Update supervisor fully developed use case description

5.3 Activity Diagrams

5.3.1 LOGIN

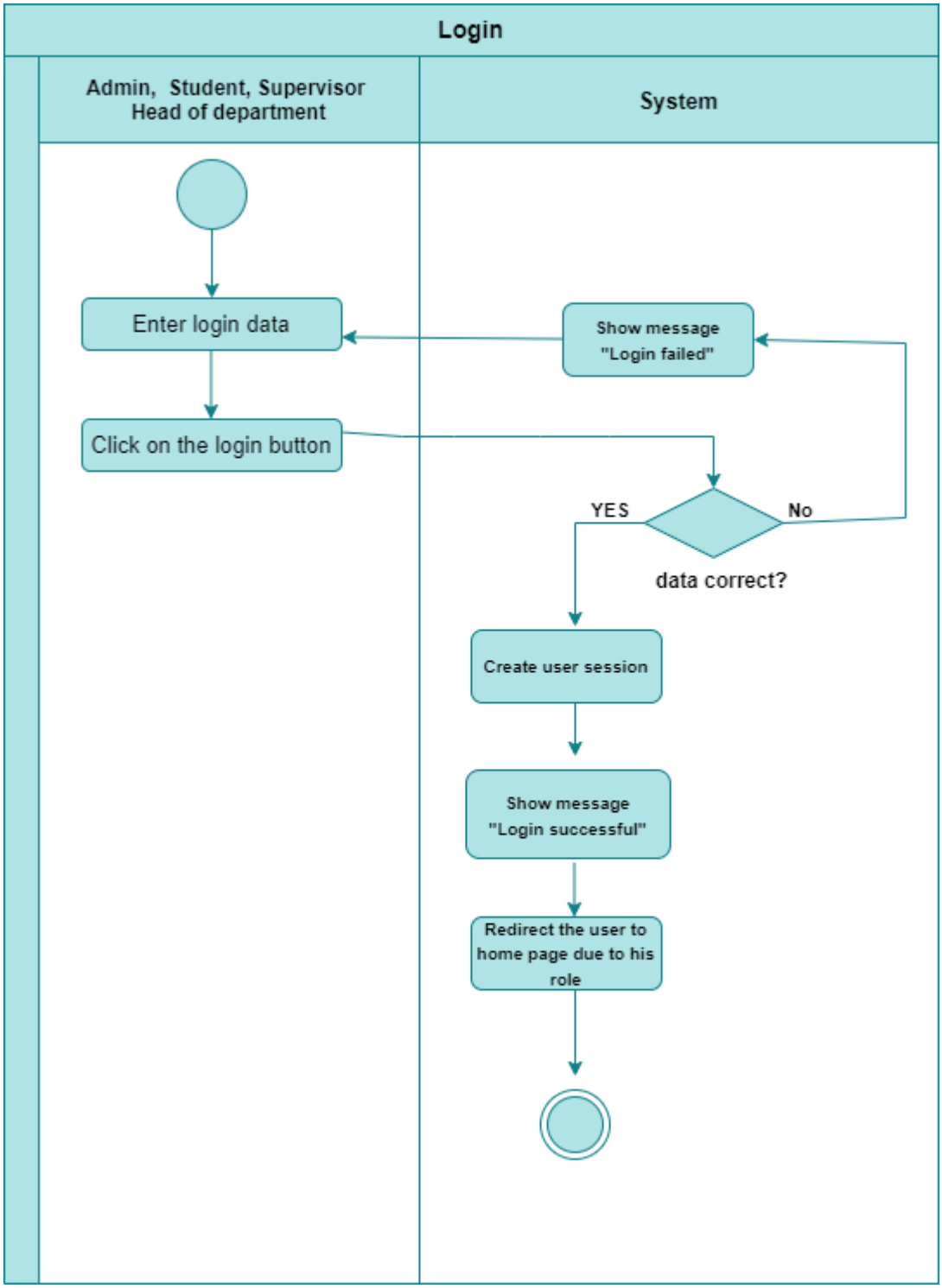

Figure 20: Login activity diagram

5.3.2 Delete Project

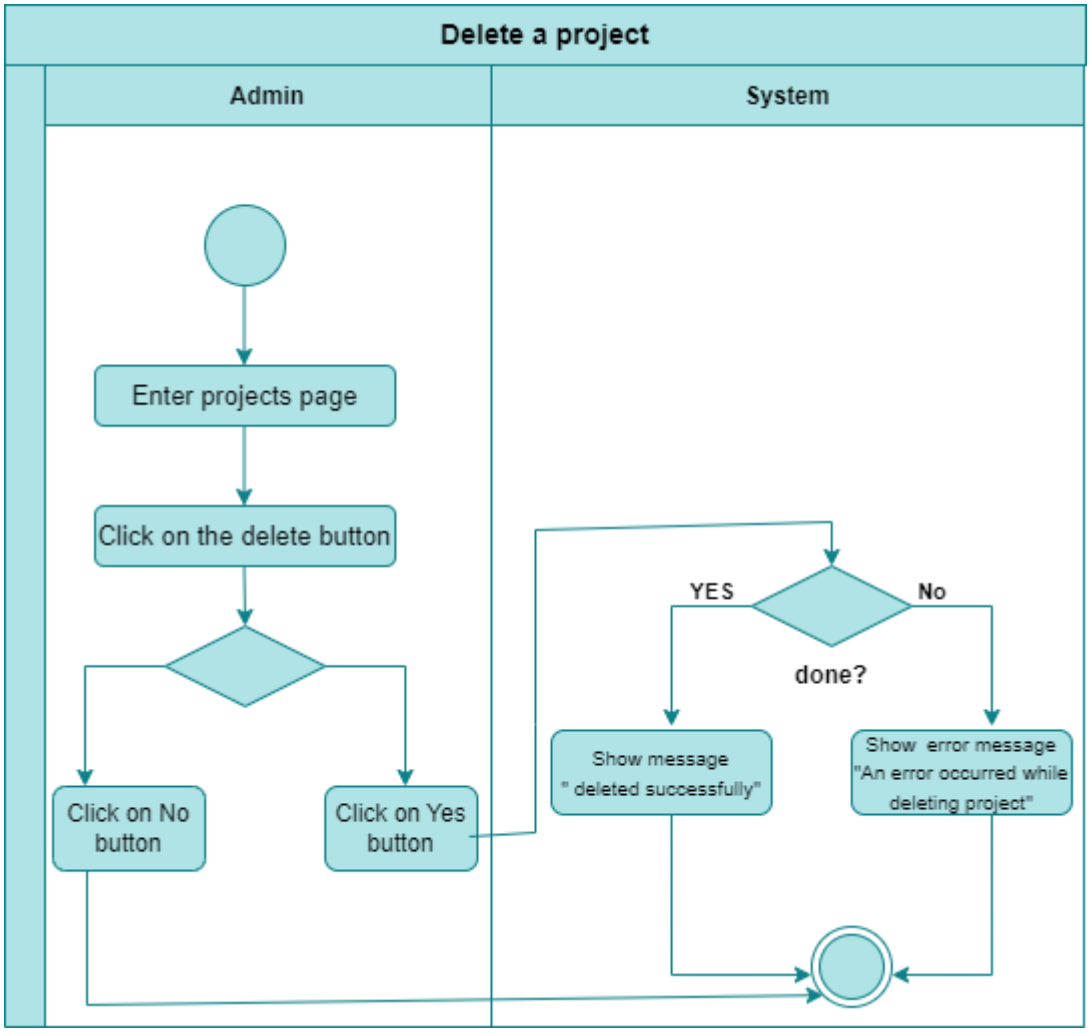

Figure 21: Delete project activity diagram

5.3.3 Add Task

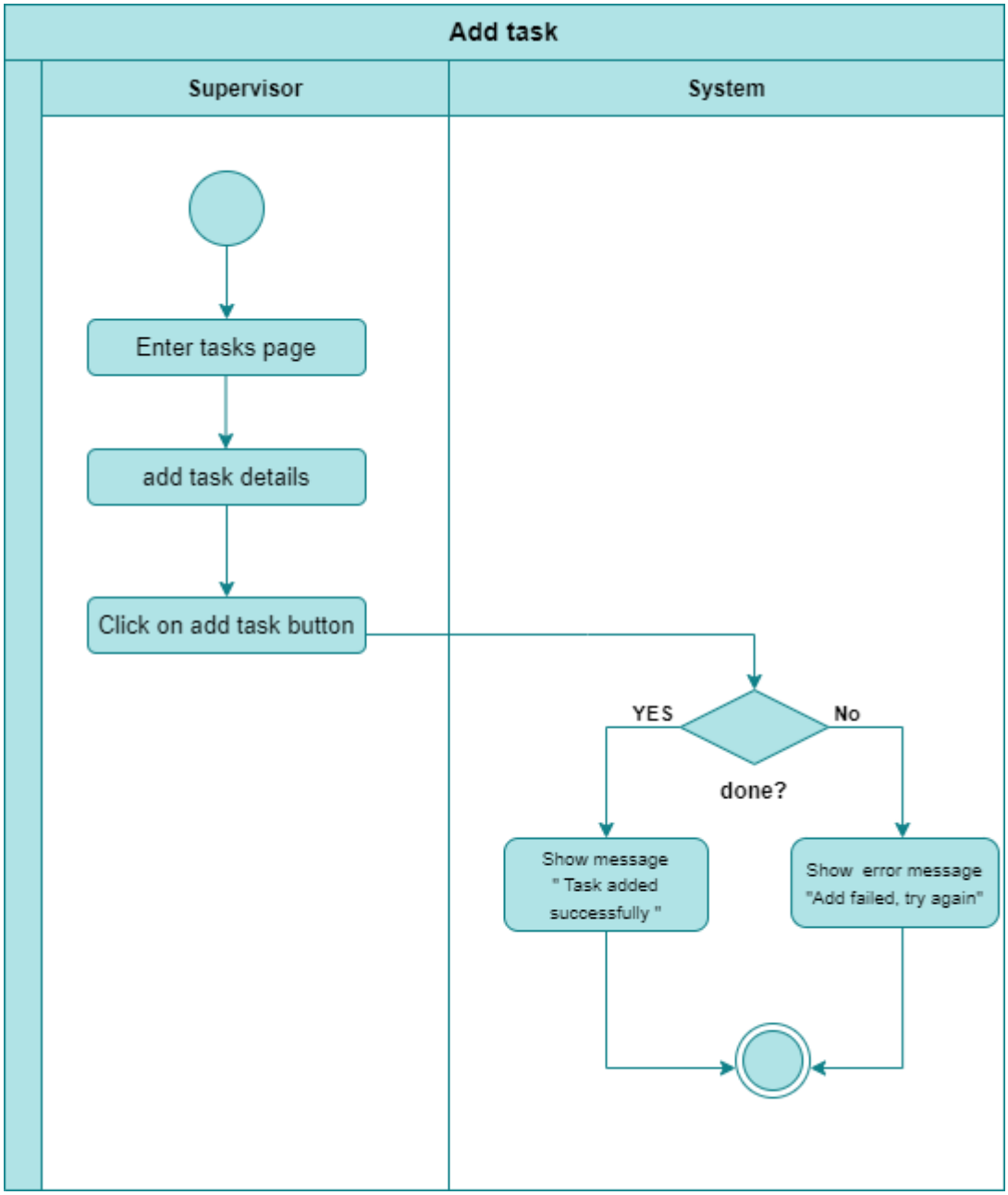

Figure 22: Add task activity diagram

5.3.4 Submit Task

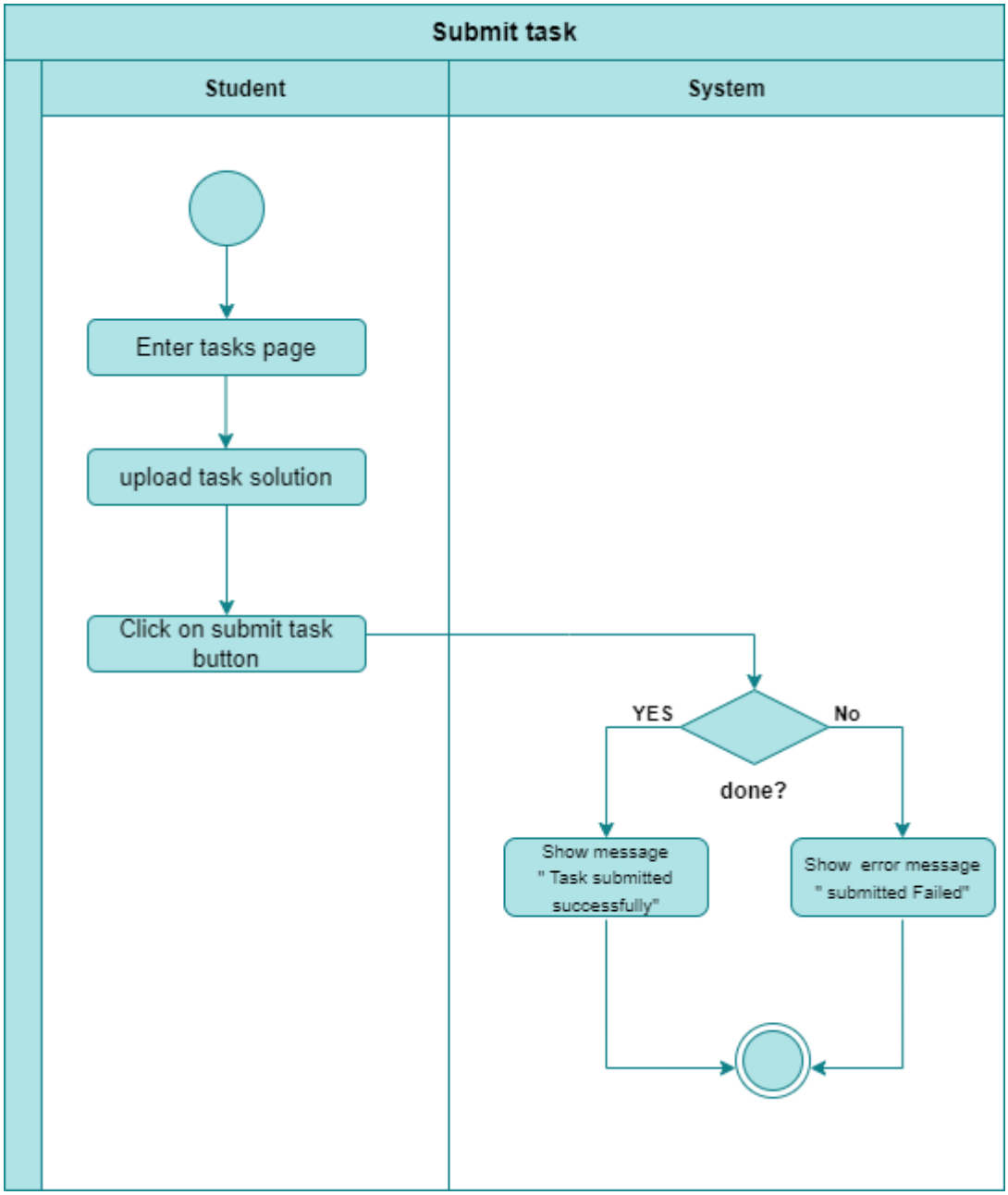

Figure 23: Submit task activity diagram

5.4 Sequence Diagrams

5.4.1 Login

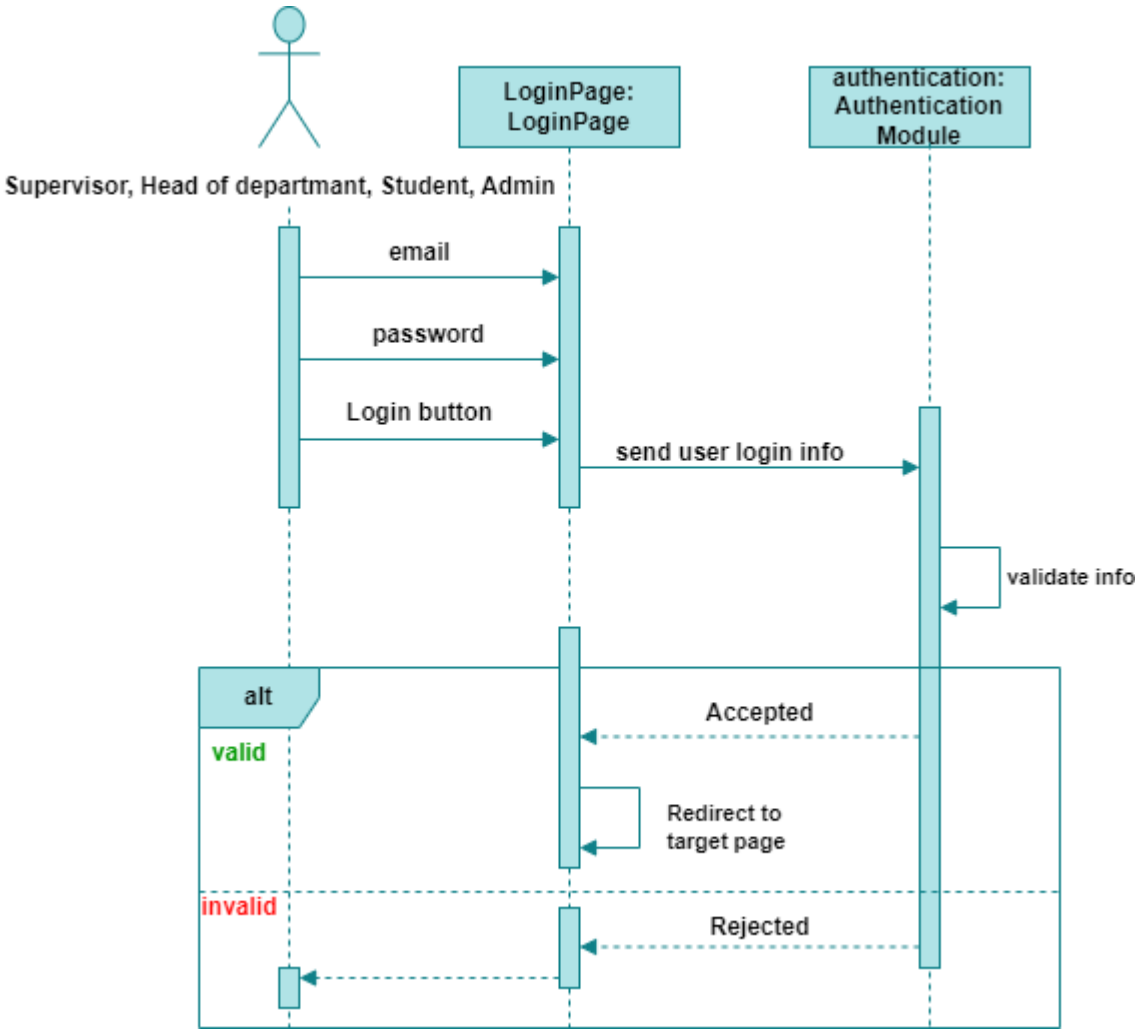

Figure 24: Login sequence diagram

5.4.2 Logout

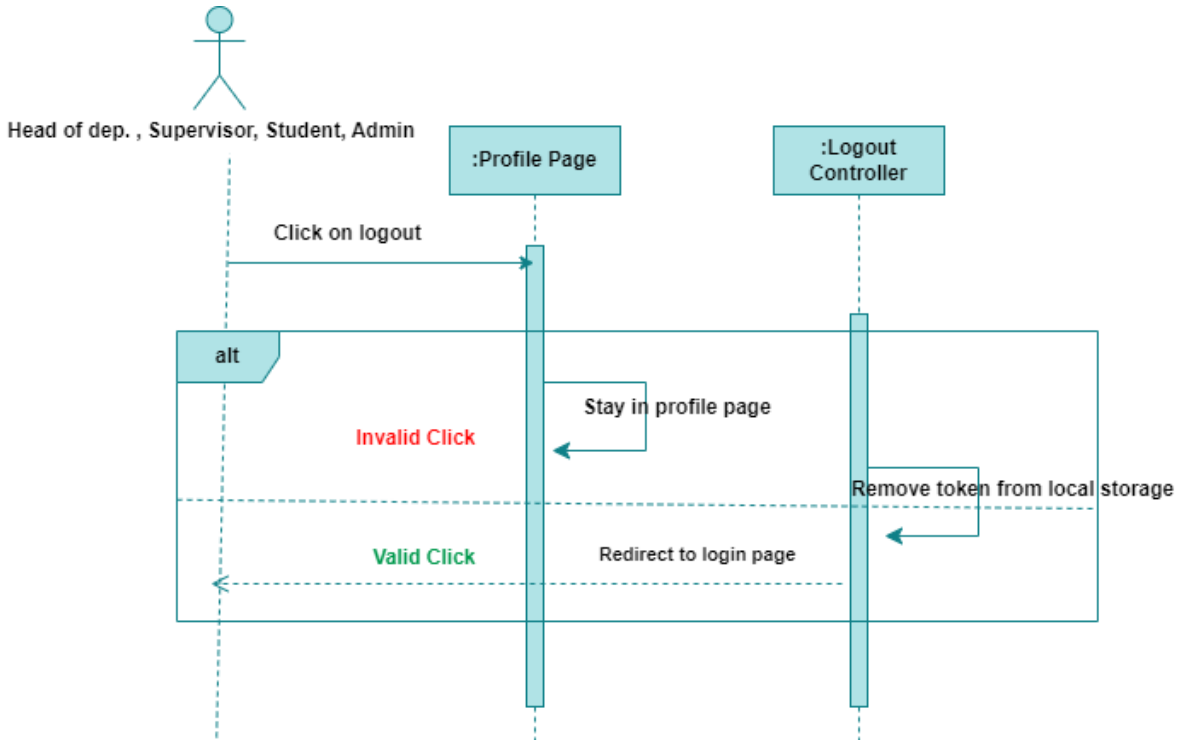

Figure 25: Logout sequence diagram

5.5 Domain Class Diagram

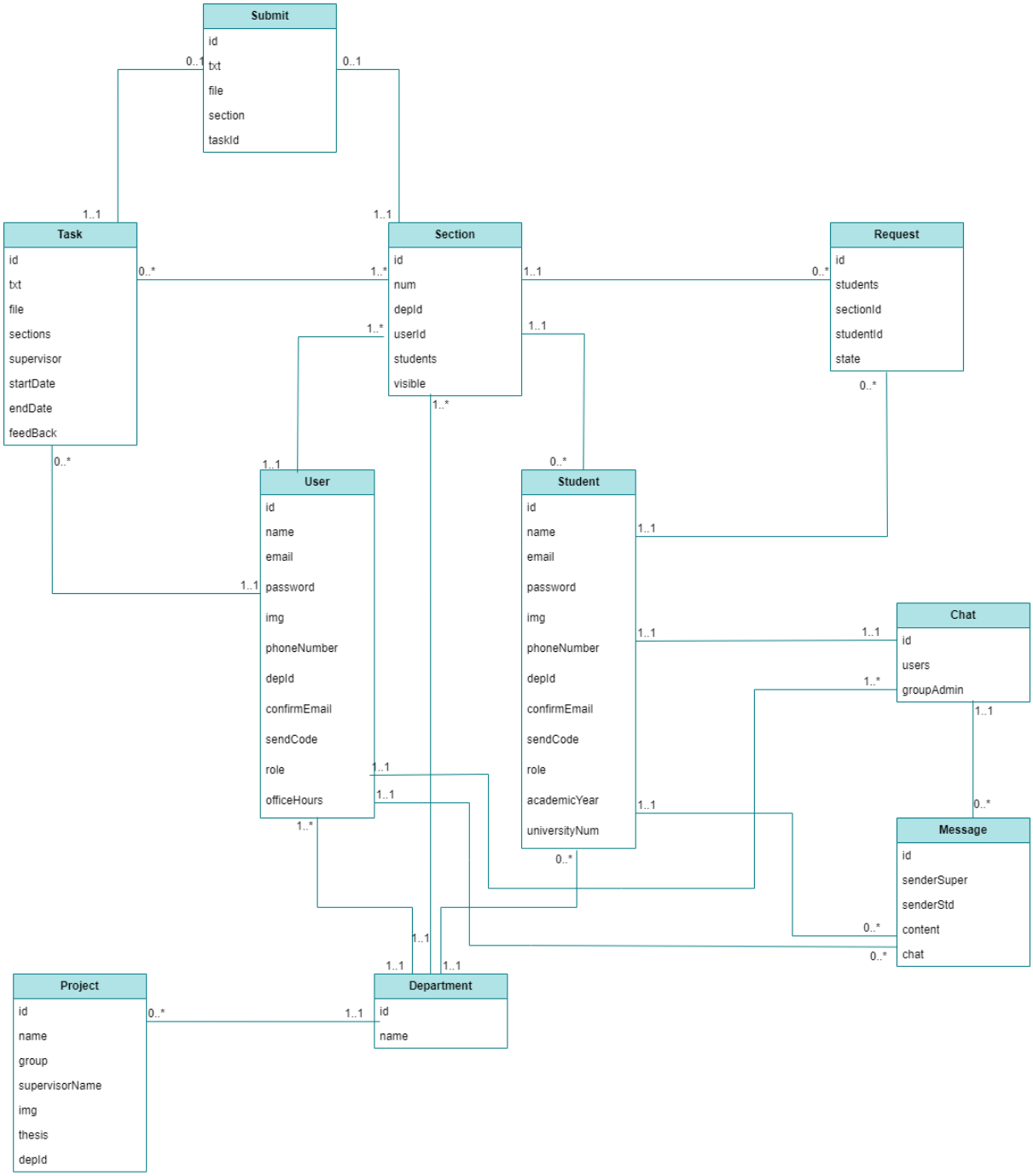

Figure 26: Domain class diagram

CHAPTER SIX

USER INTERFACE

Admin

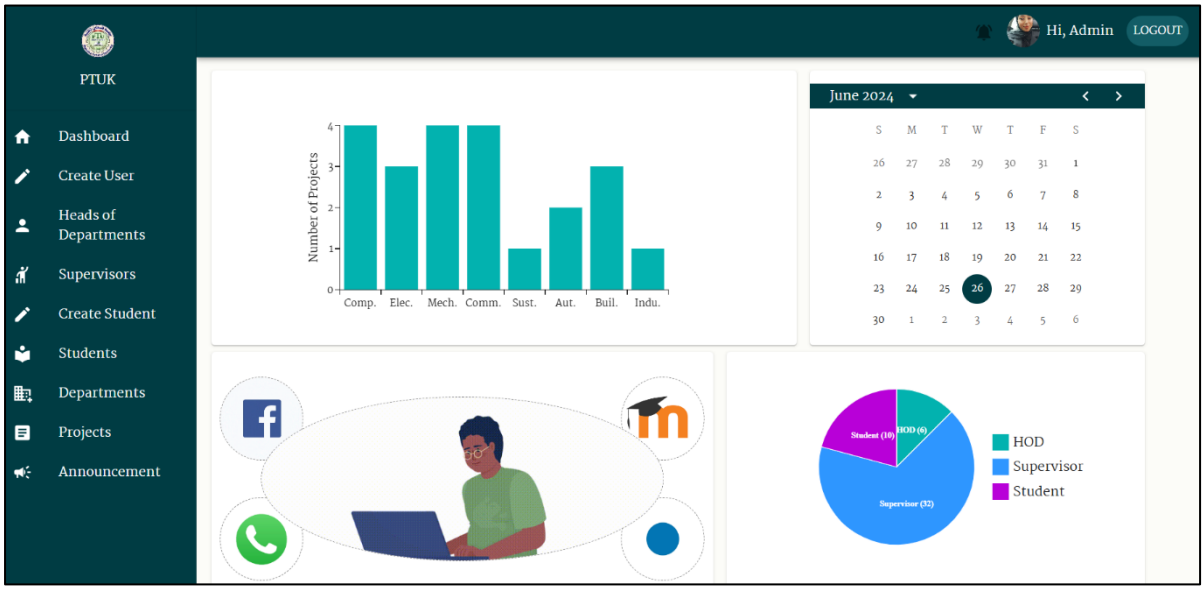

Figure 27: Admin dashboard

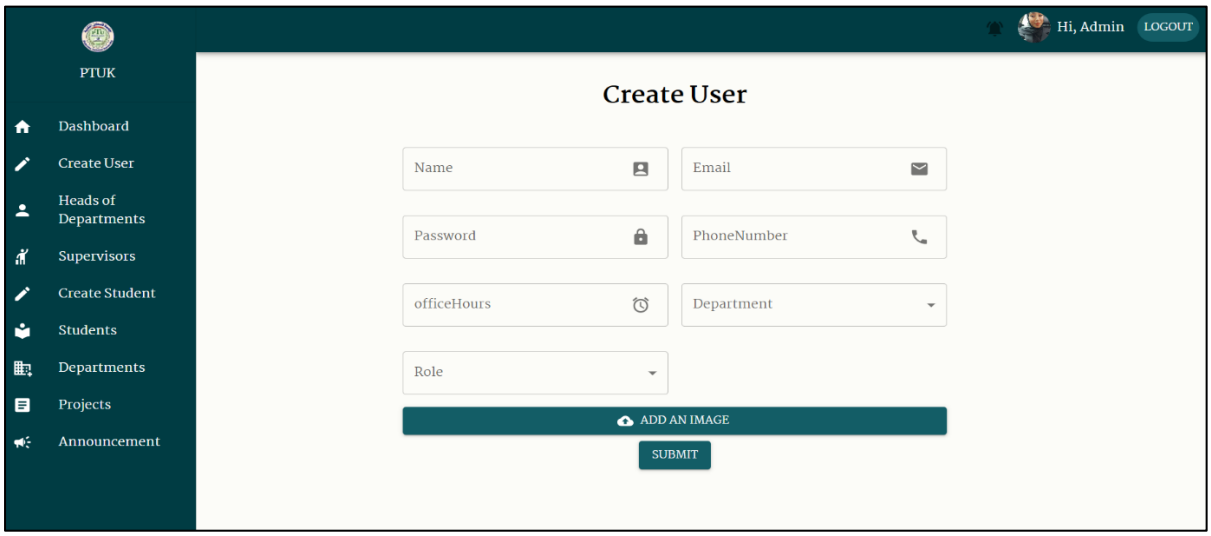

Figure 28: Create user

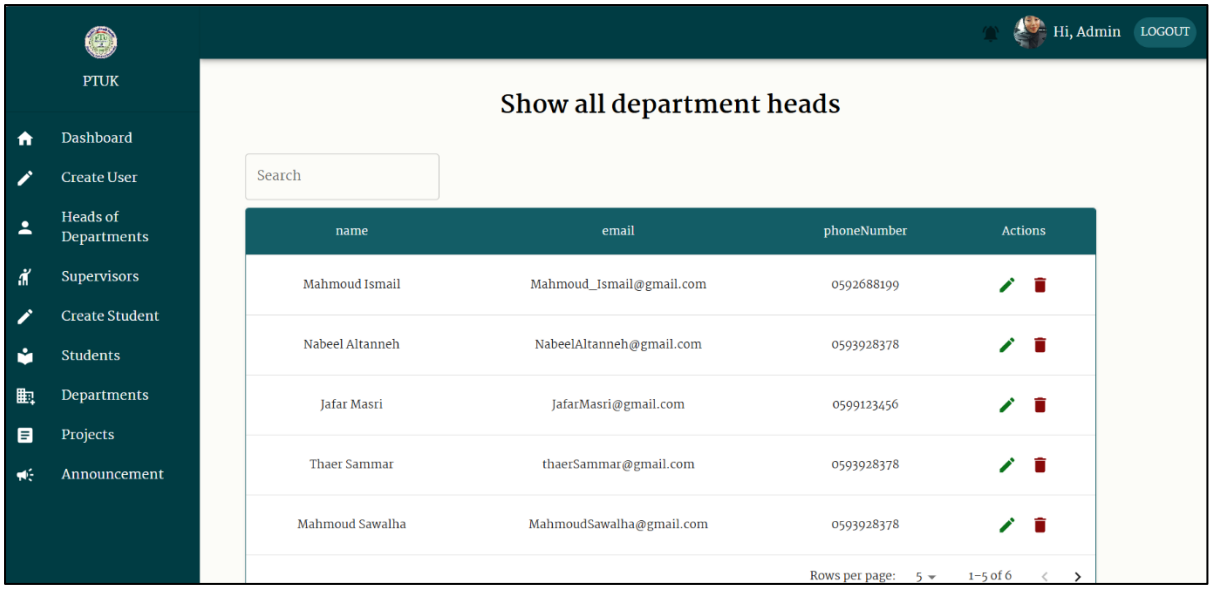

Figure 29: Display department heads

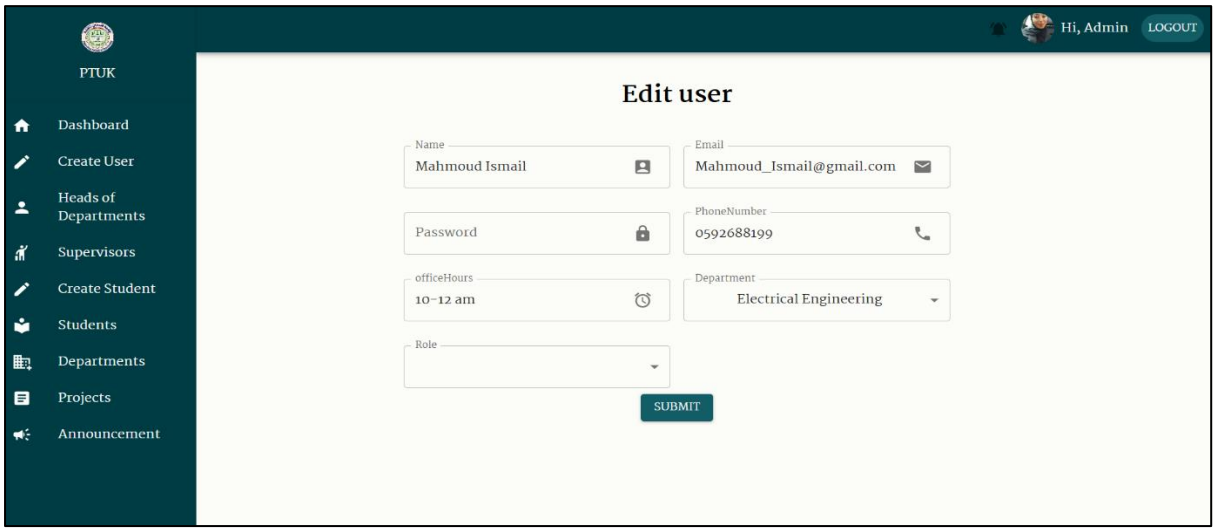

Figure 30: Edit user

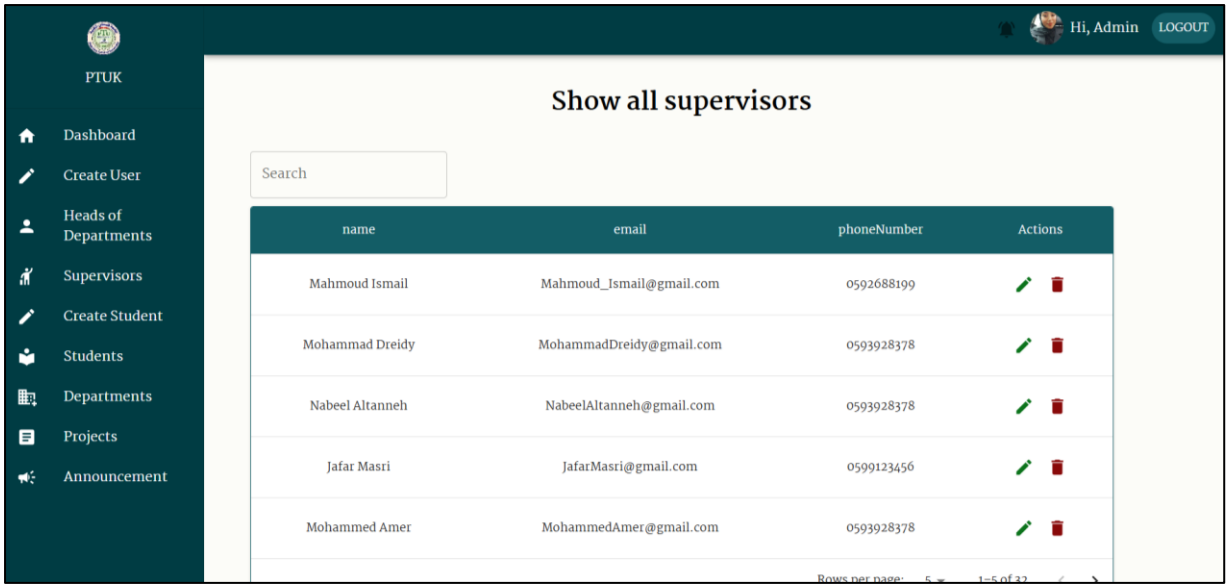

Figure 31: Display supervisors

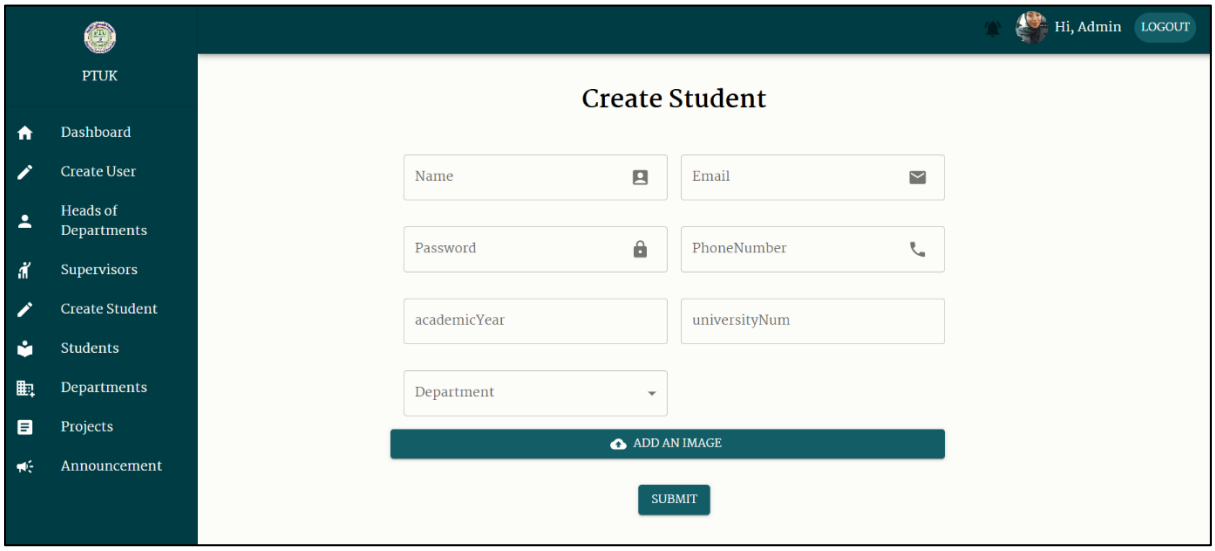

Figure 32: Create student

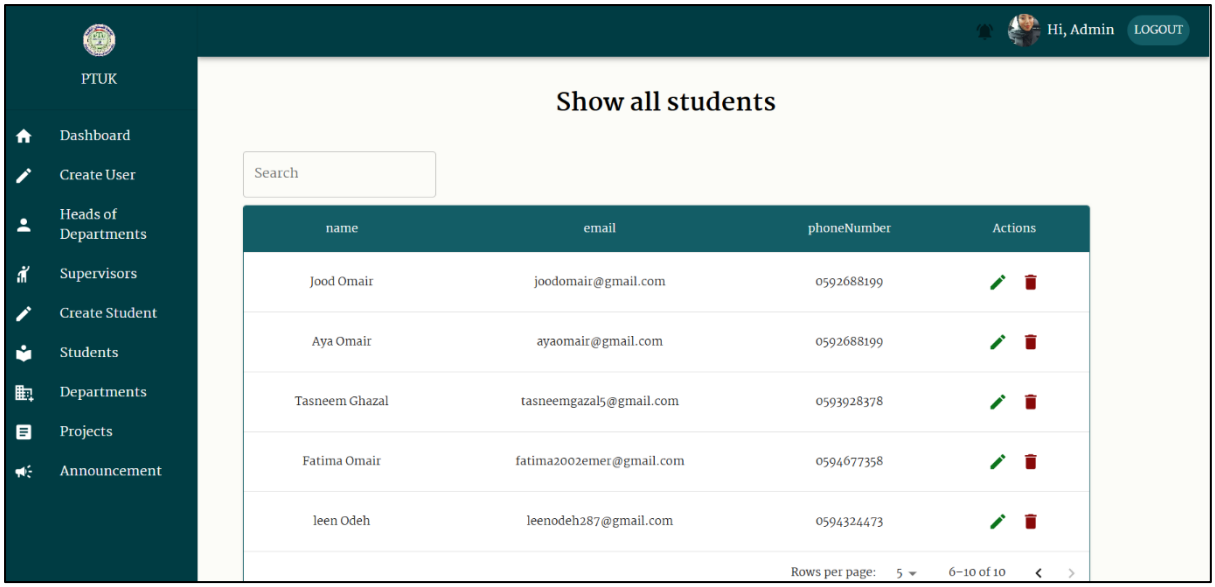

Figure 33: Display all students

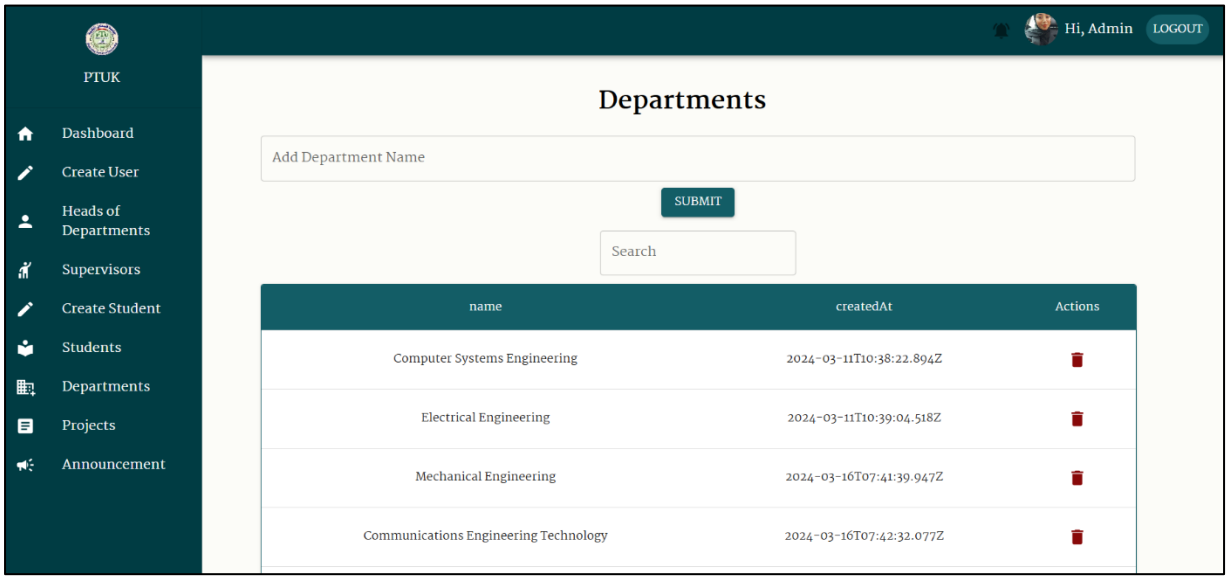

Figure 34: Add and display departments

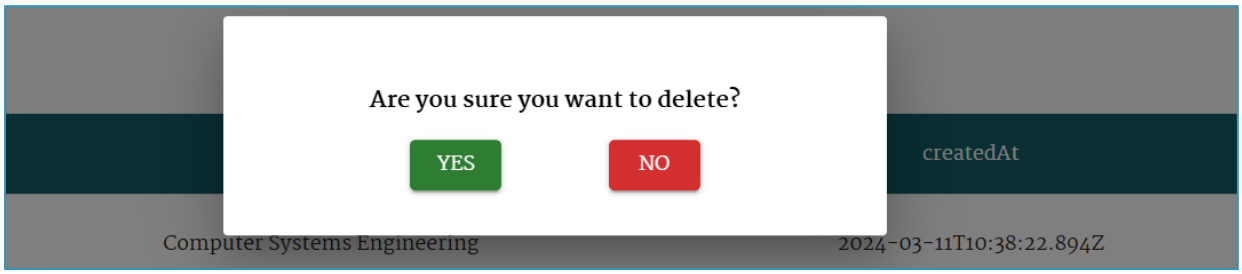

Figure 35: Confirm delete modal

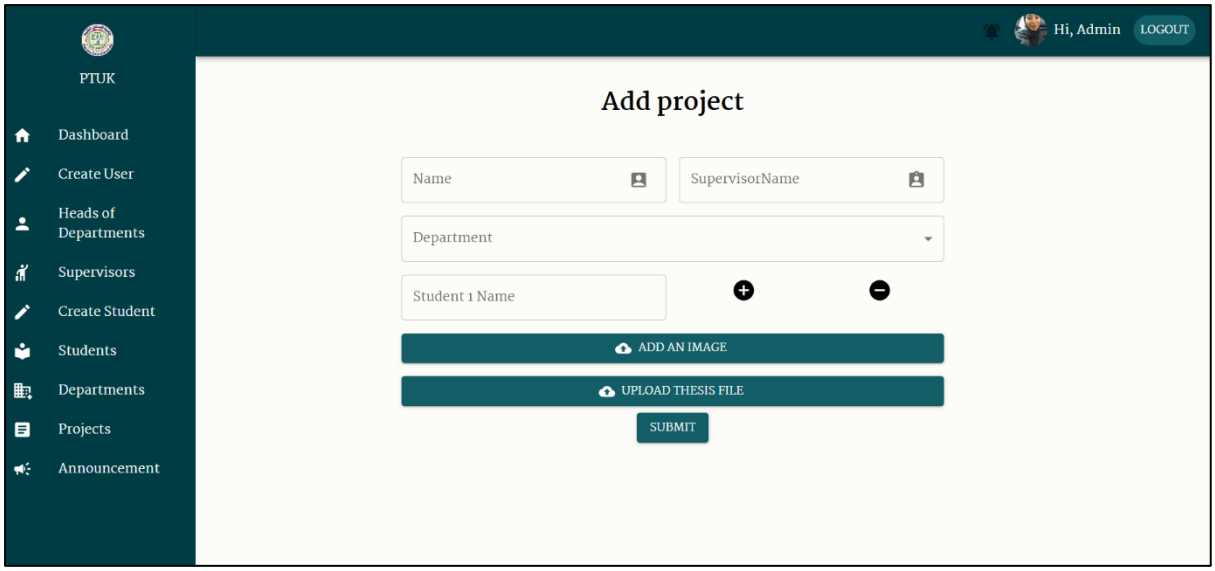

Figure 36: Add project

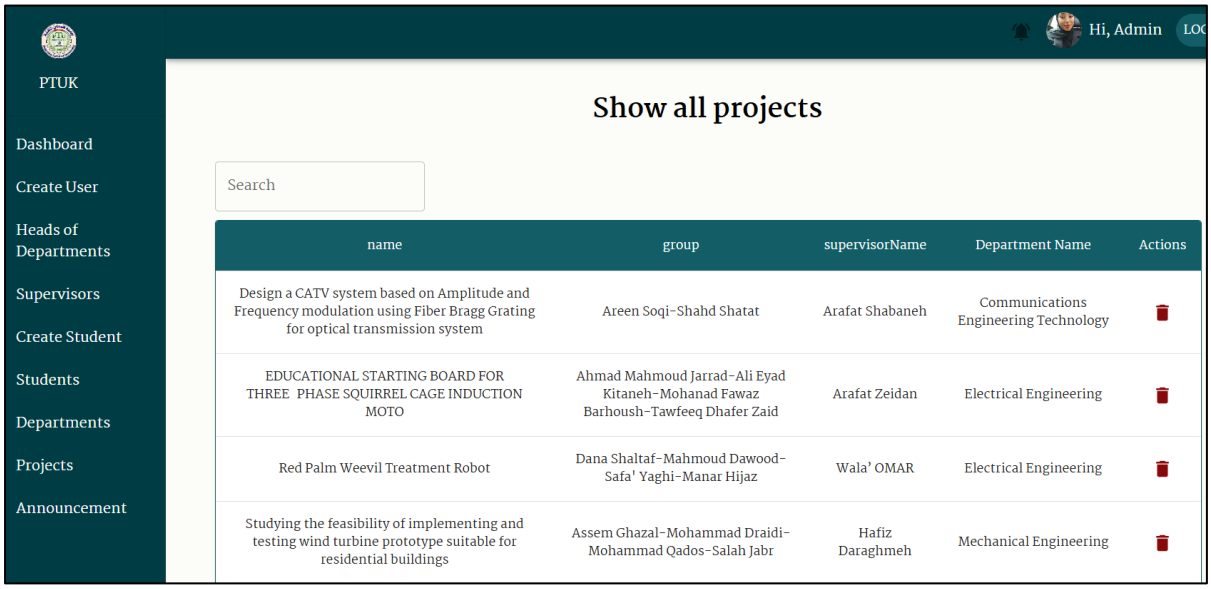

Figure 37: Display projects

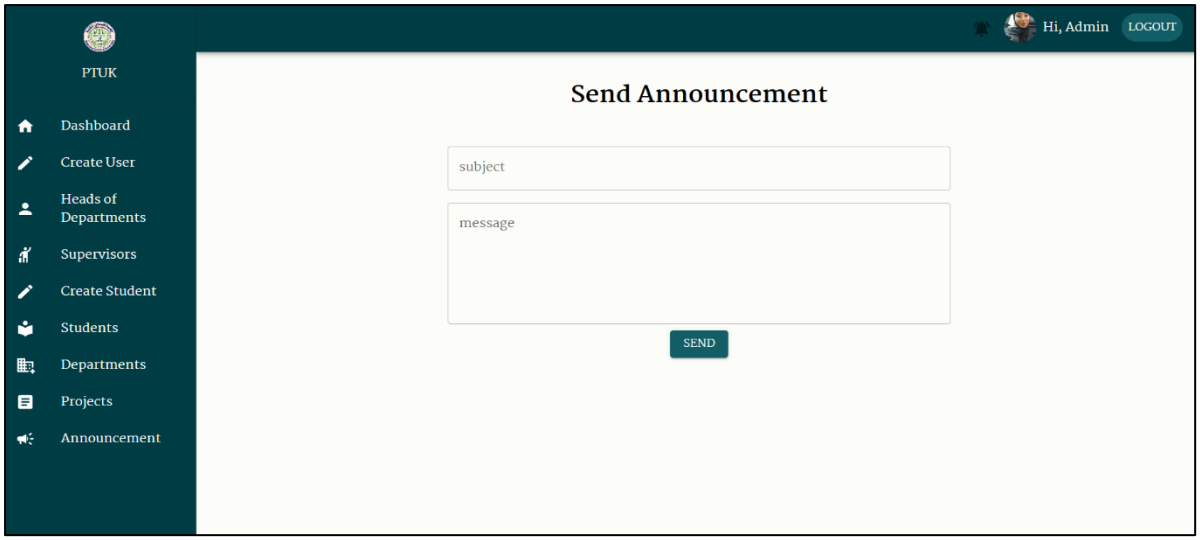

Figure 38: Send announcement

Main Page

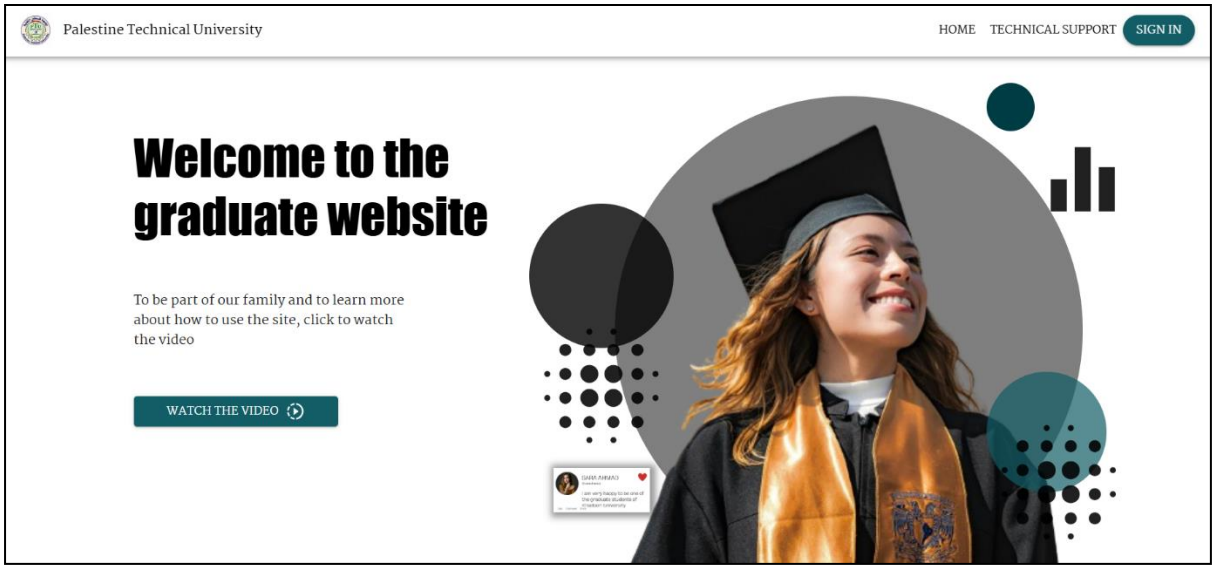

Figure 39: Main page header

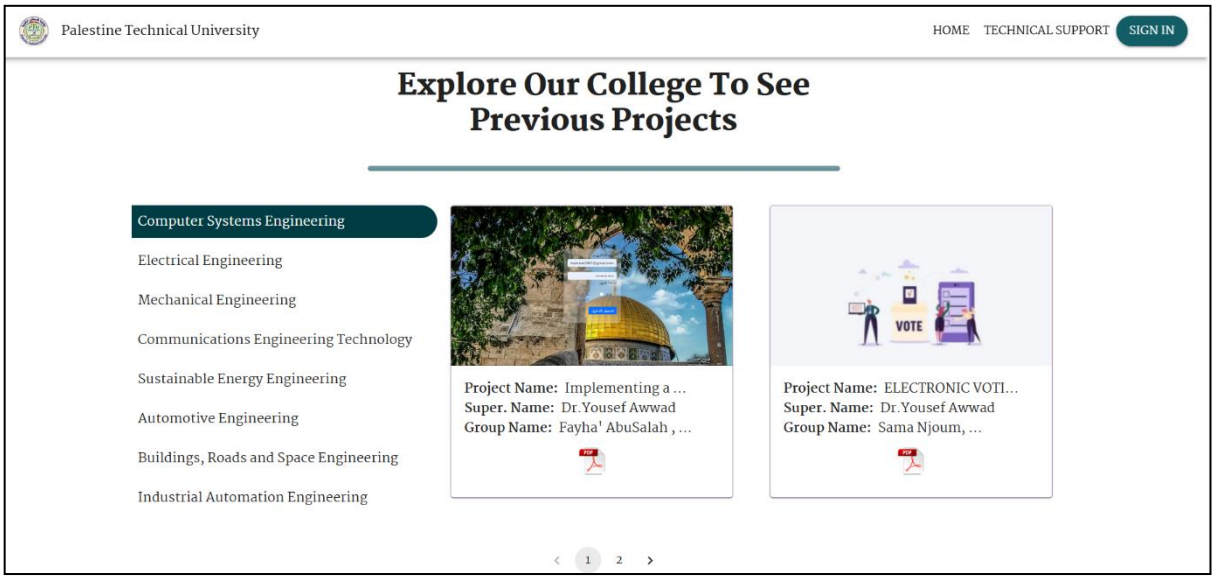

Figure 40: Previous projects section

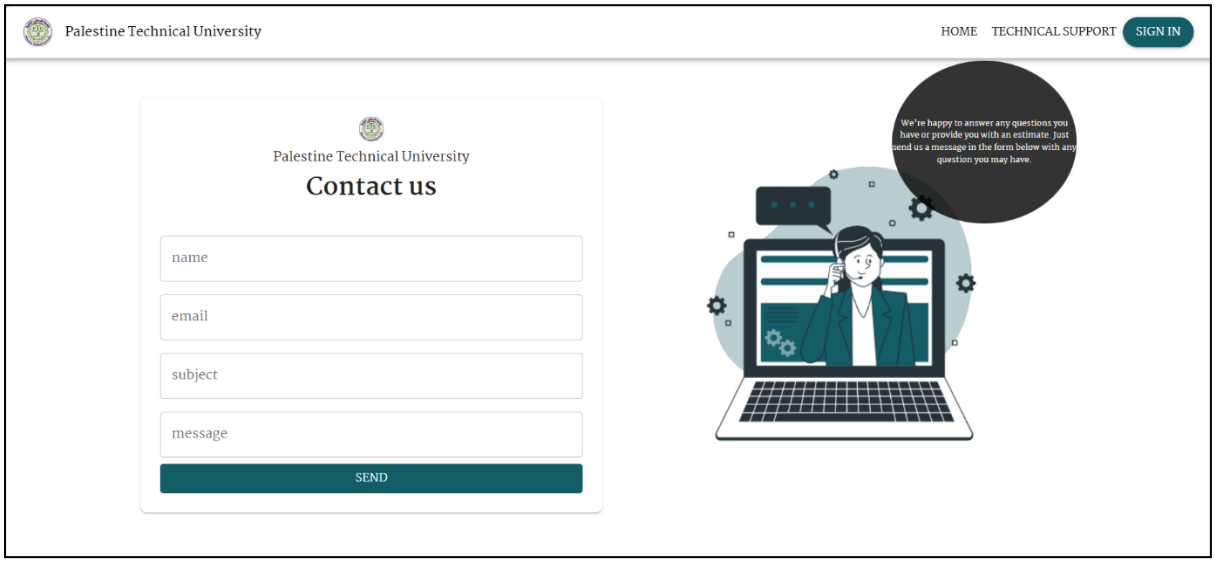

Figure 41: Technical support

Sign In Page

Figure 42: Sign in

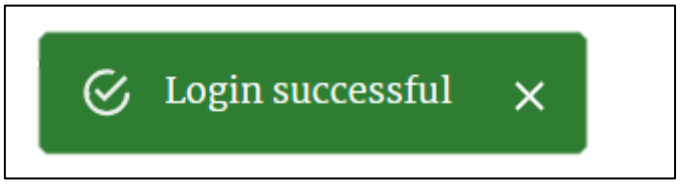

Figure 43: Login success snake bar

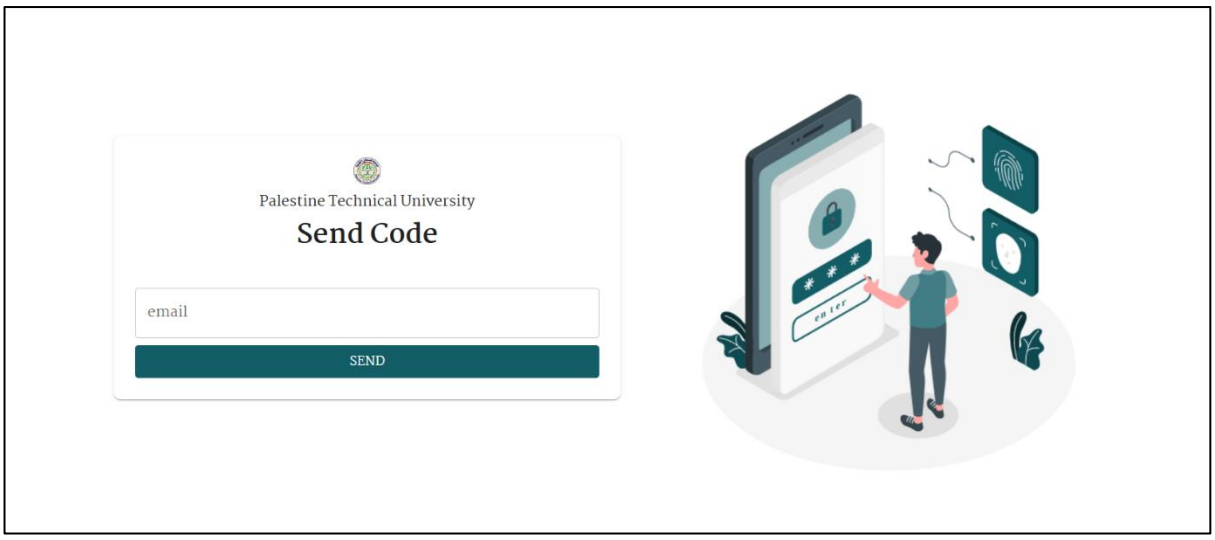

Figure 44: Send Code

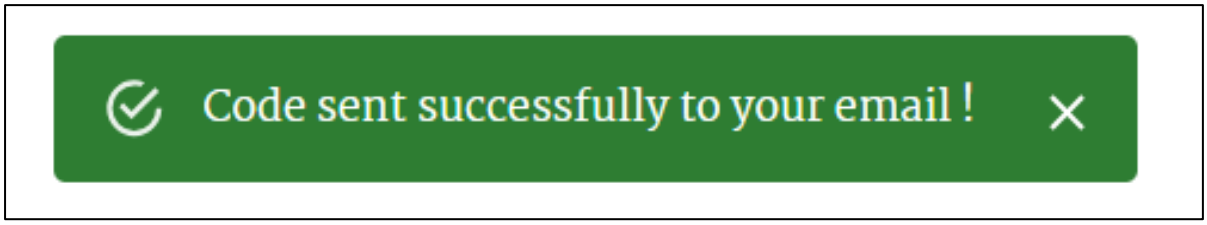

Figure 45: send code success snake bar

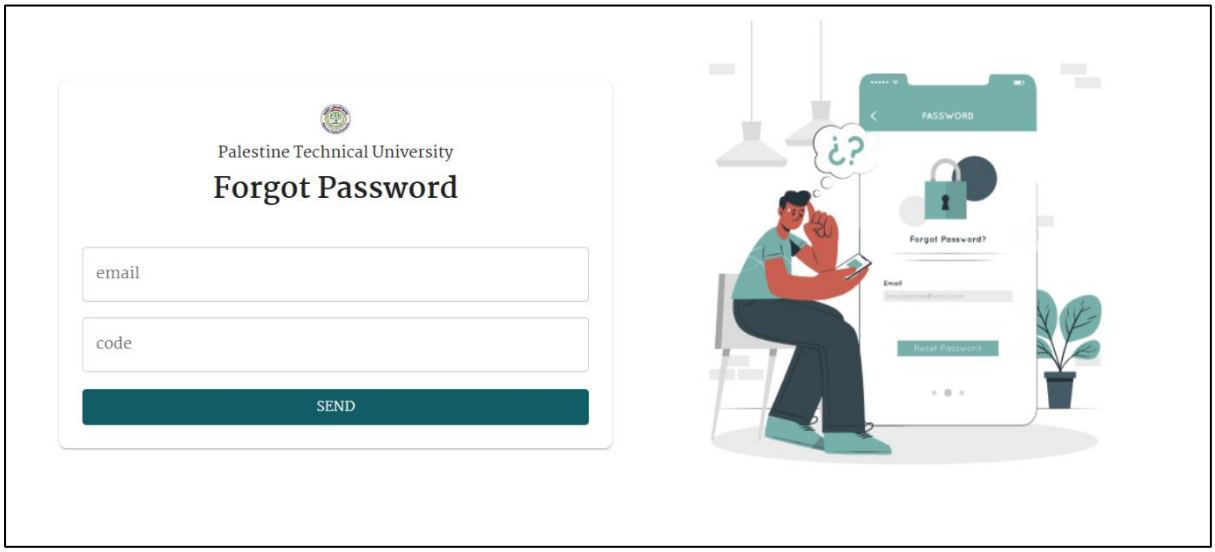

Figure 46: Forget password

Student Page

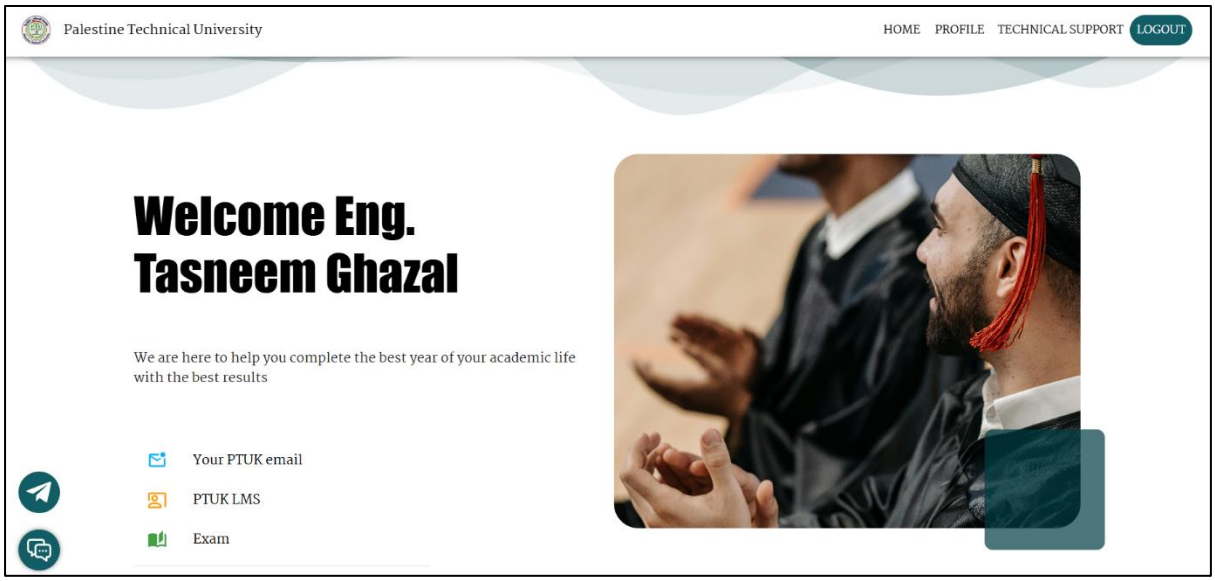

Figure 47: Student page header

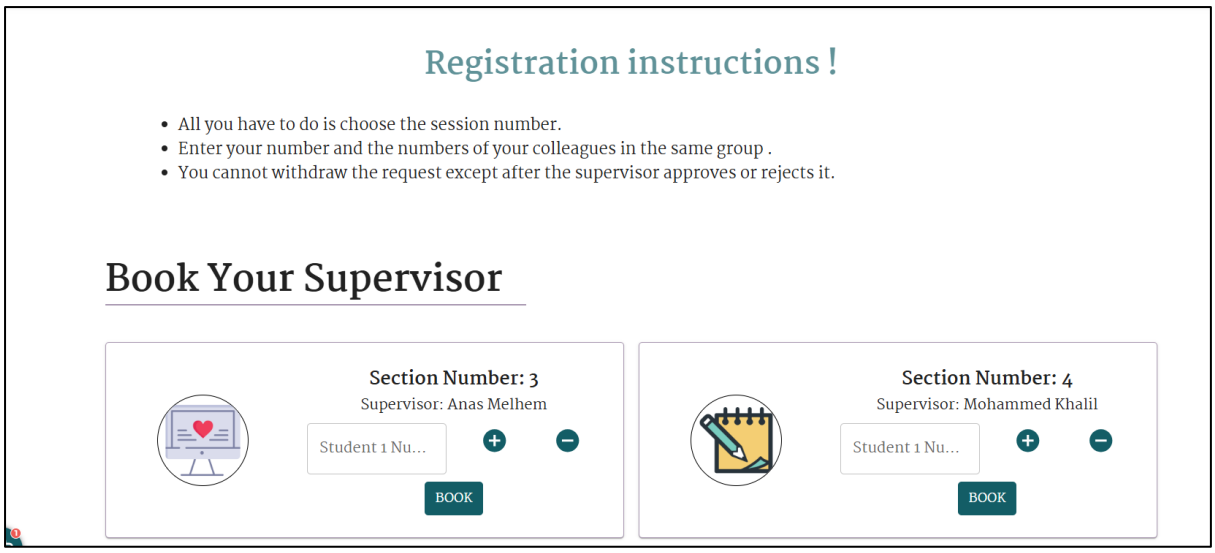

Figure 48: Book section

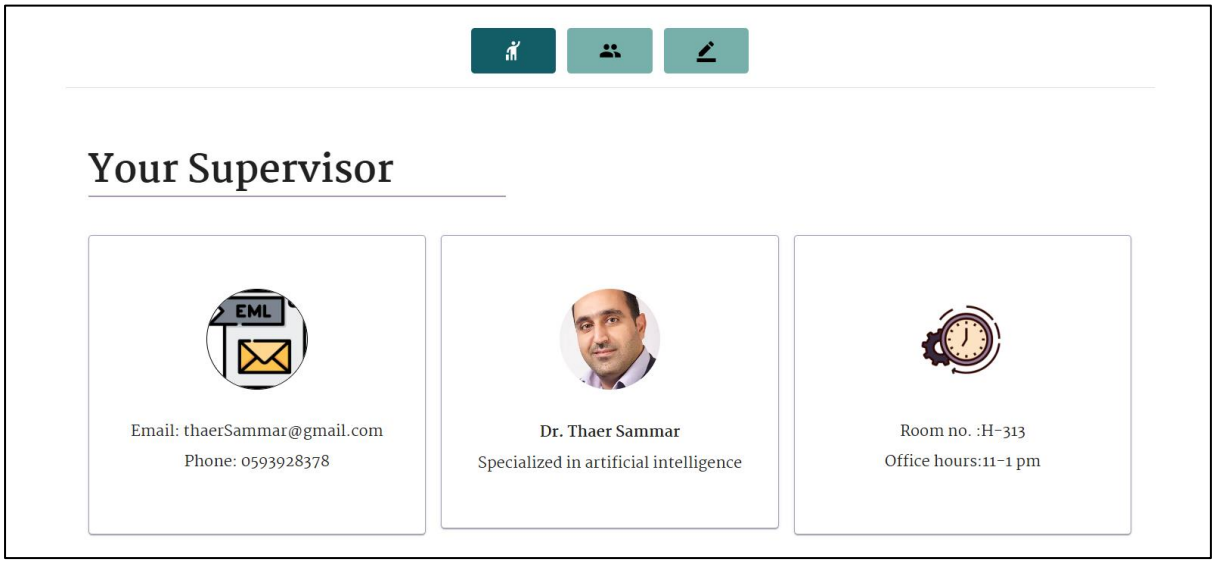

Figure 49: Supervisor details tab

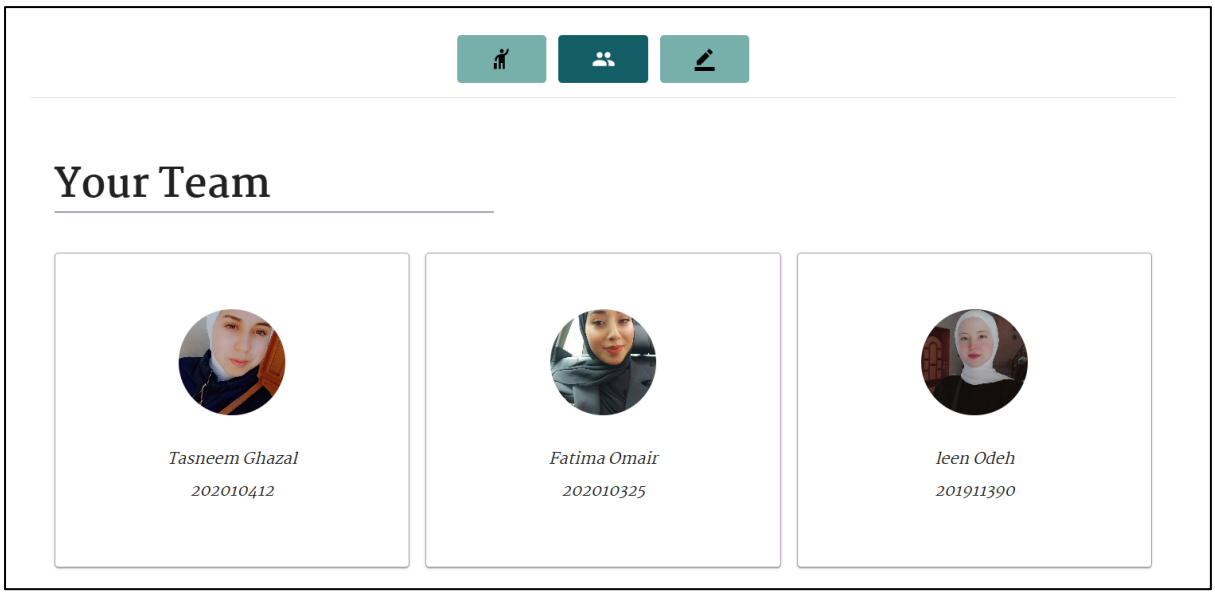

Figure 50: Team information tab

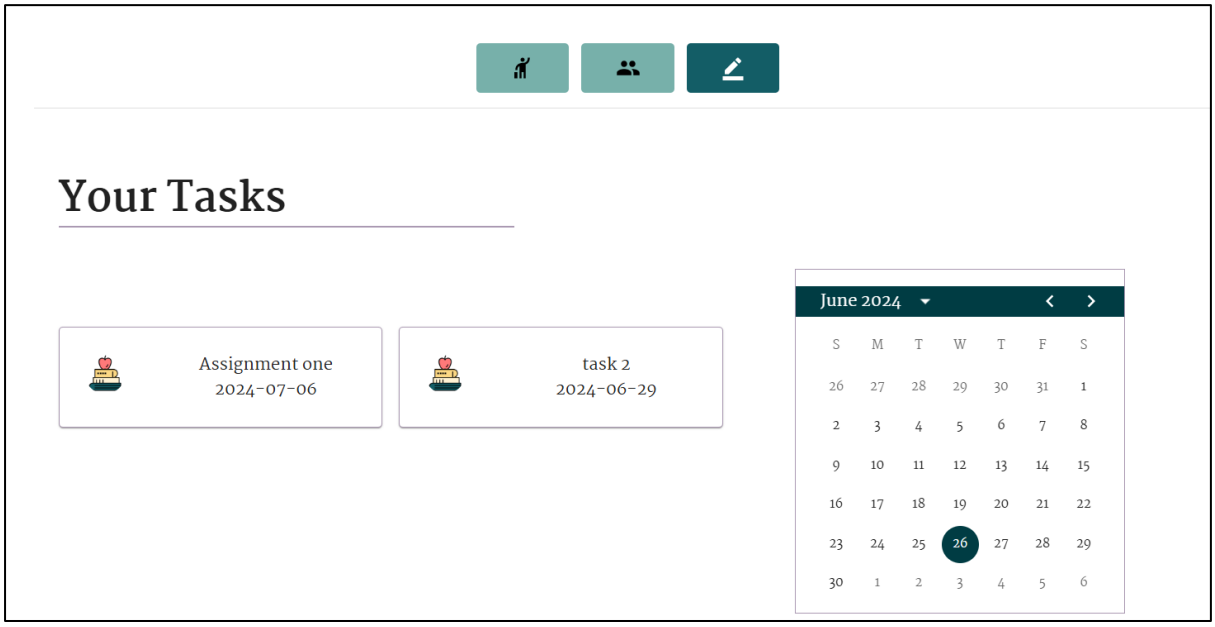

Figure 51: View tasks tab

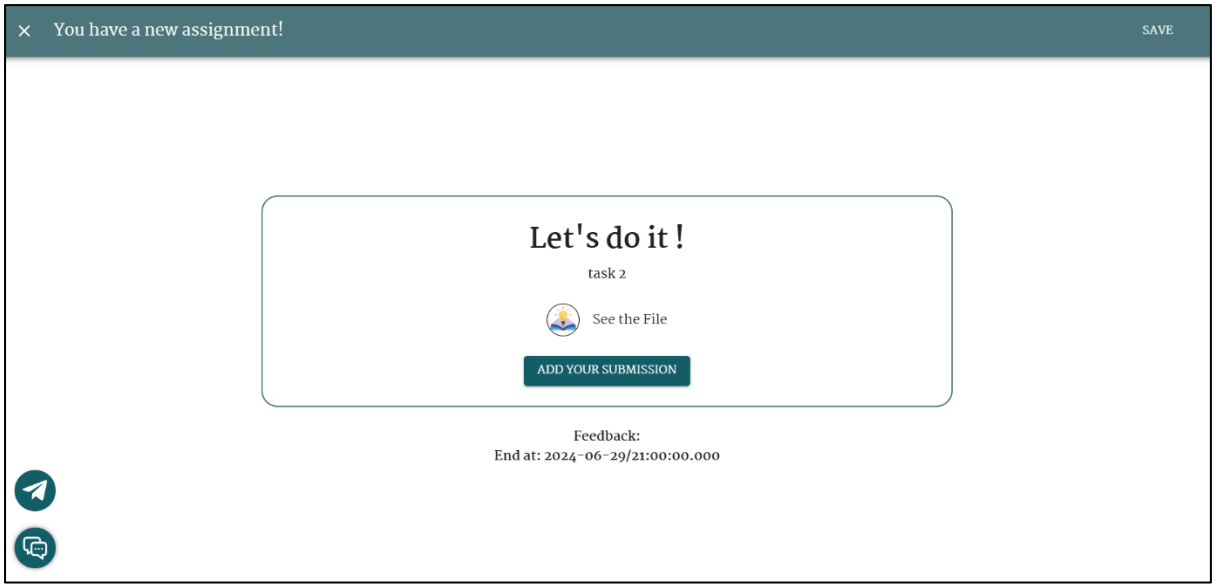

Figure 52: Task details

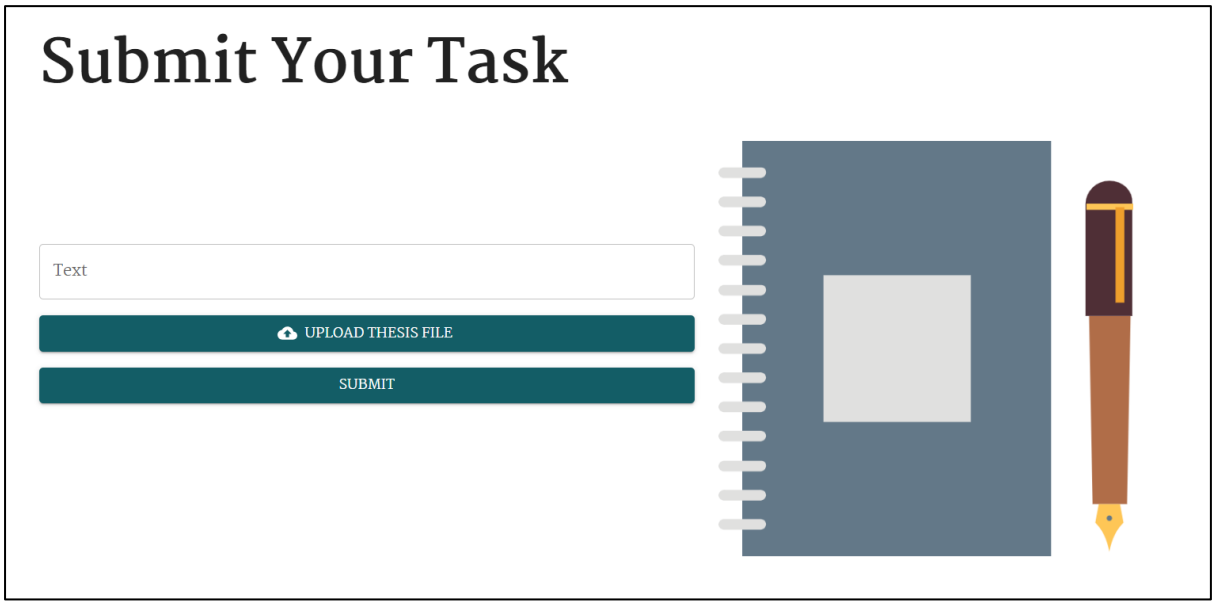

Figure 53: Submit task

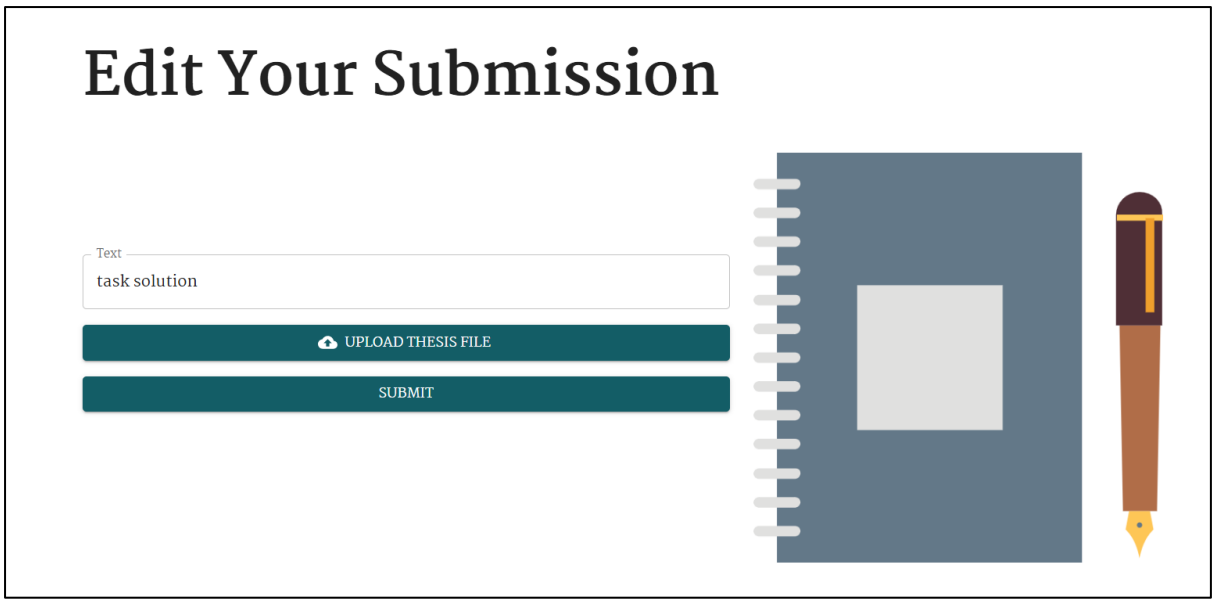

Figure 54: Edit submission

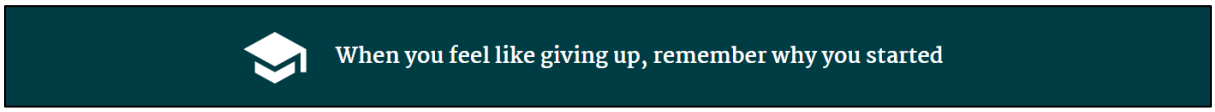

Figure 55: Student page footer

Figure 56: Student chat

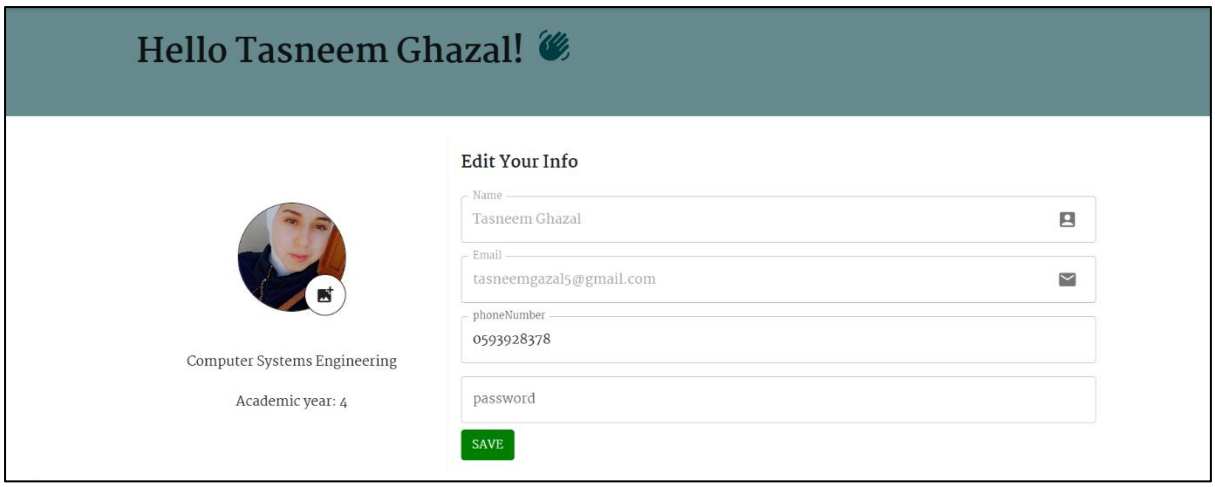

Figure 57: Student profile

Chatbot

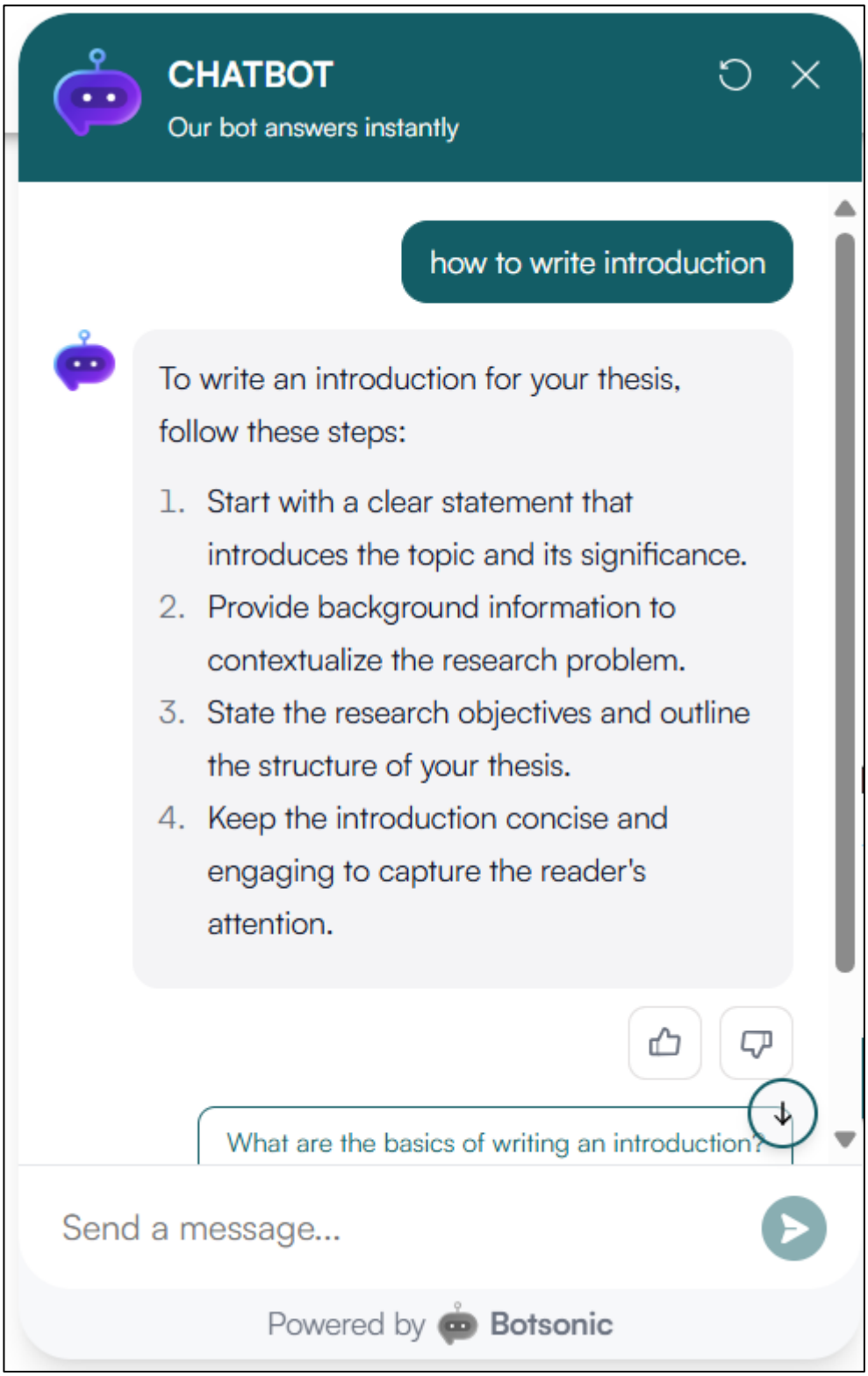

Figure 58: Chat bot

Supervisor

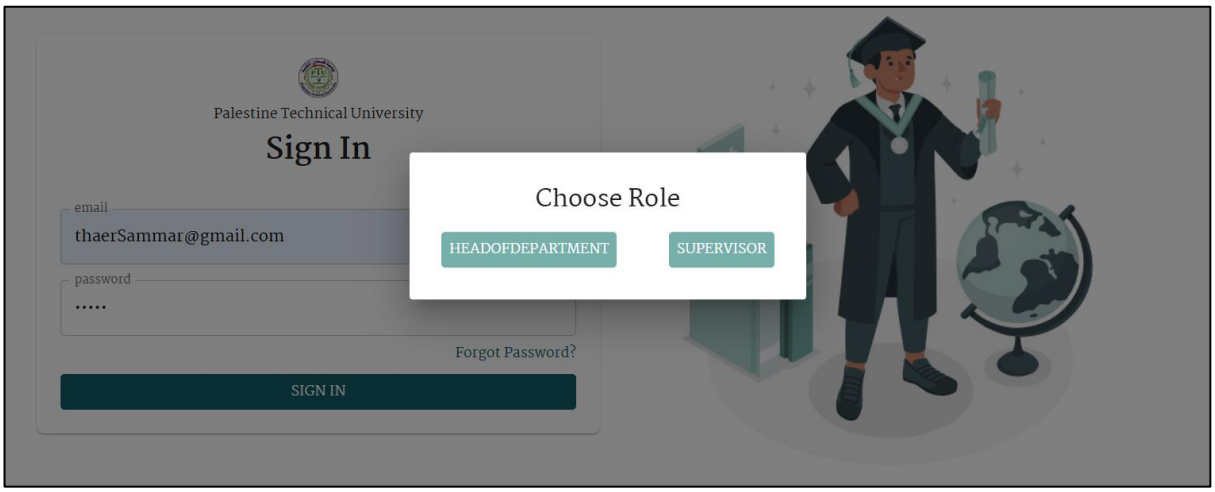

Figure 59: User with two roles modal selection

Supervisor

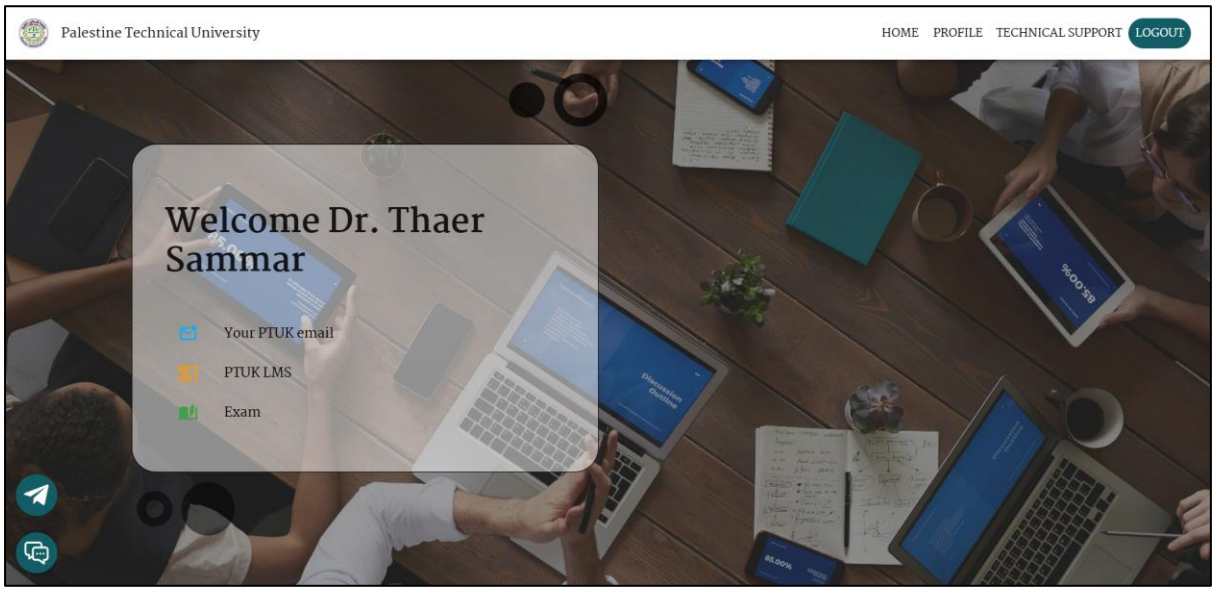

Figure 60: Supervisor page header

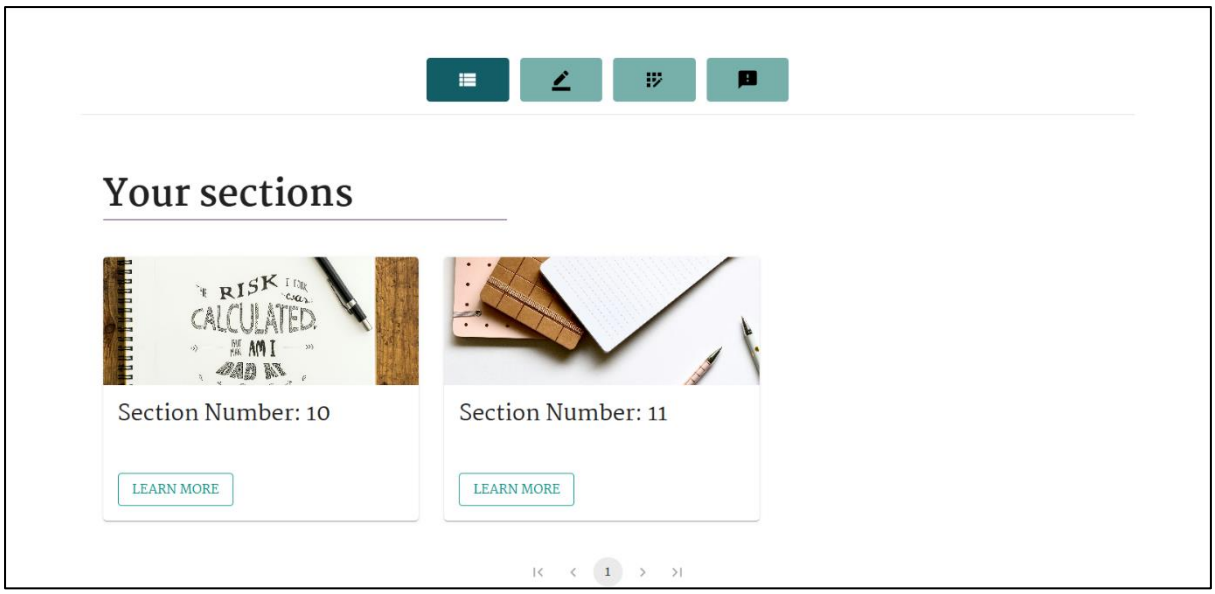

Figure 61: View sections tab

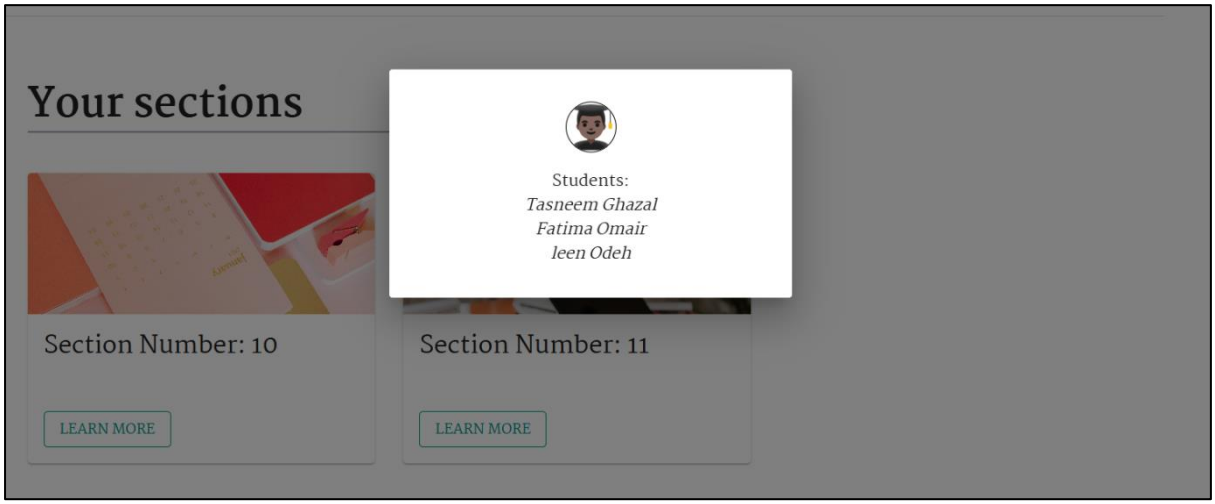

Figure 62: Students section modal

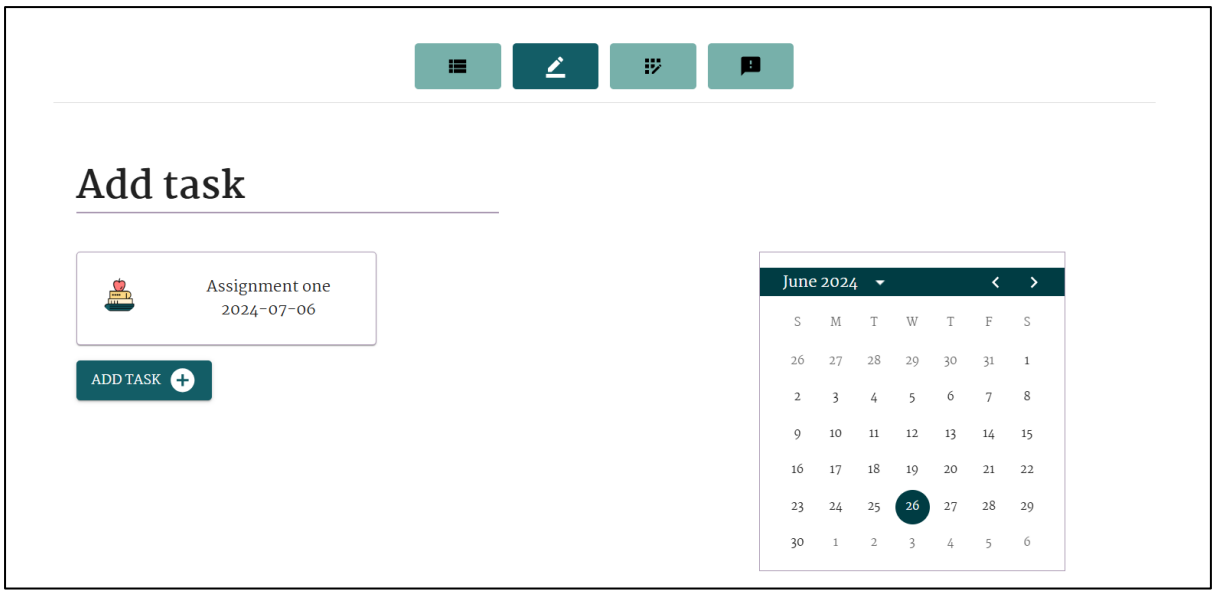

Figure 63: Add task tab

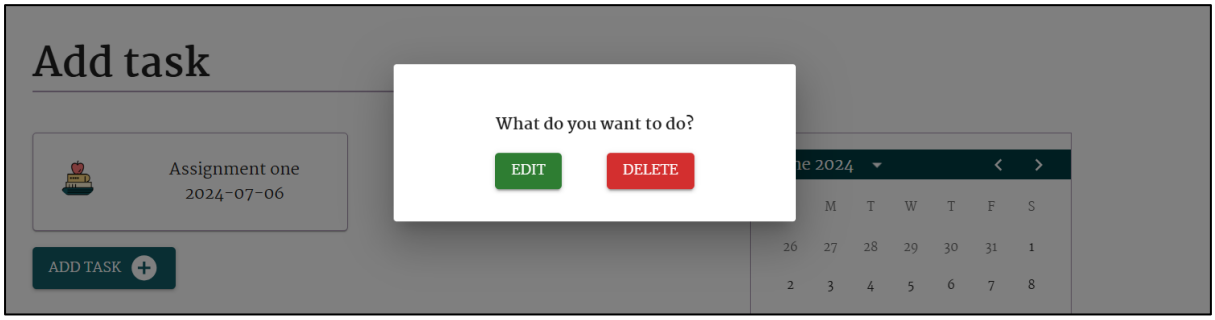

Figure 64: Edit or Delete modal

Figure 65: Edit task

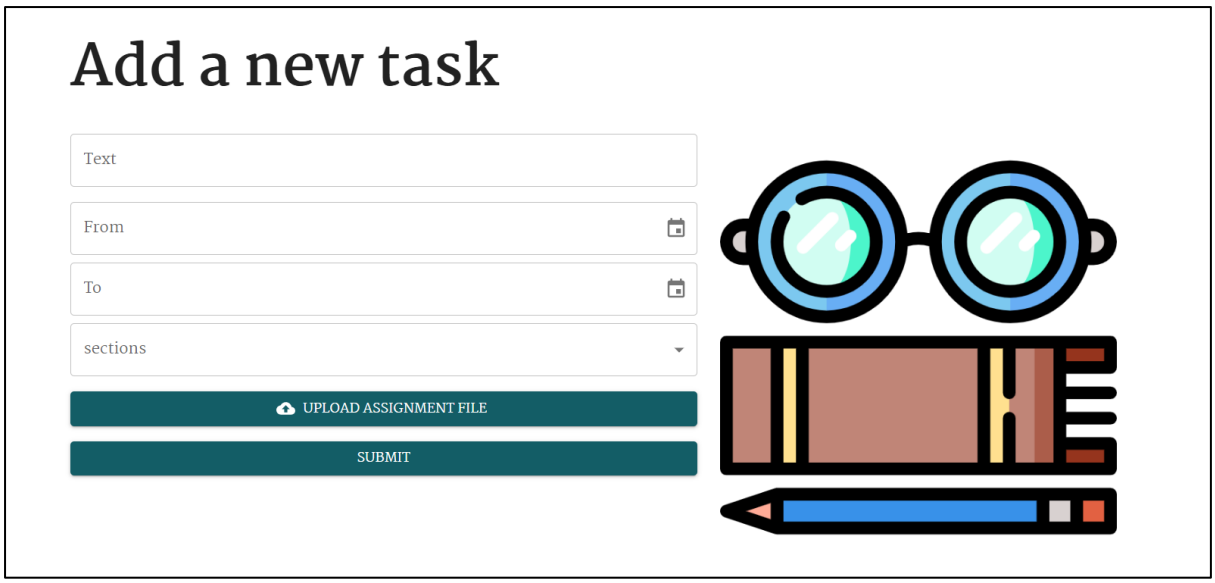

Figure 66: Add new task

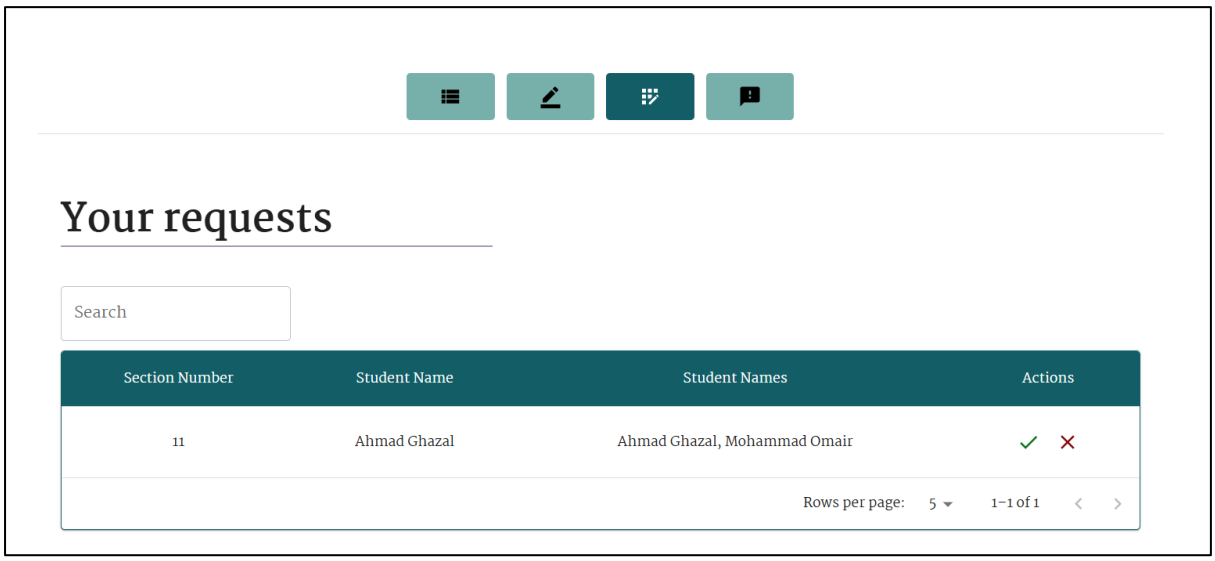

Figure 67: View requests tab

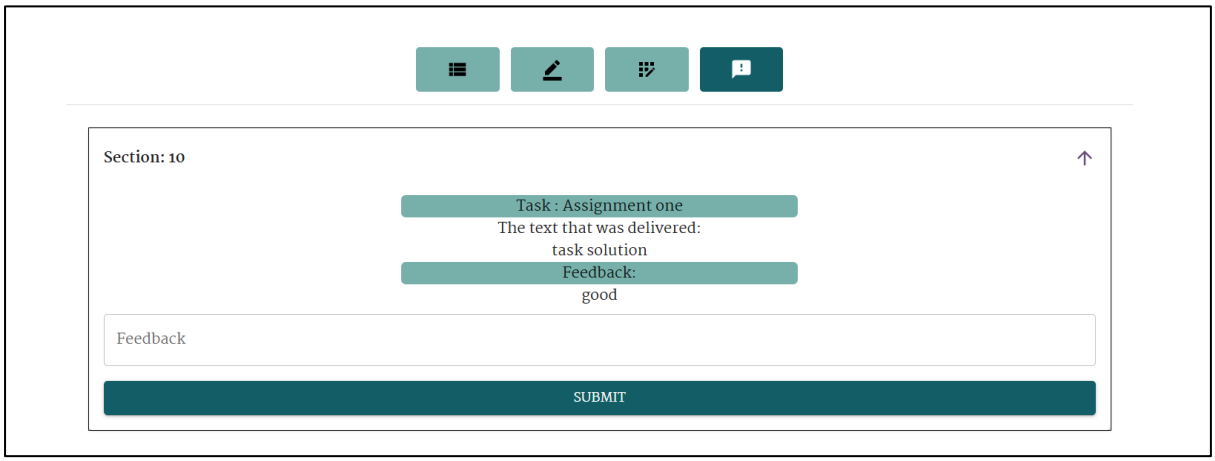

Figure 68: Give Feedback tab

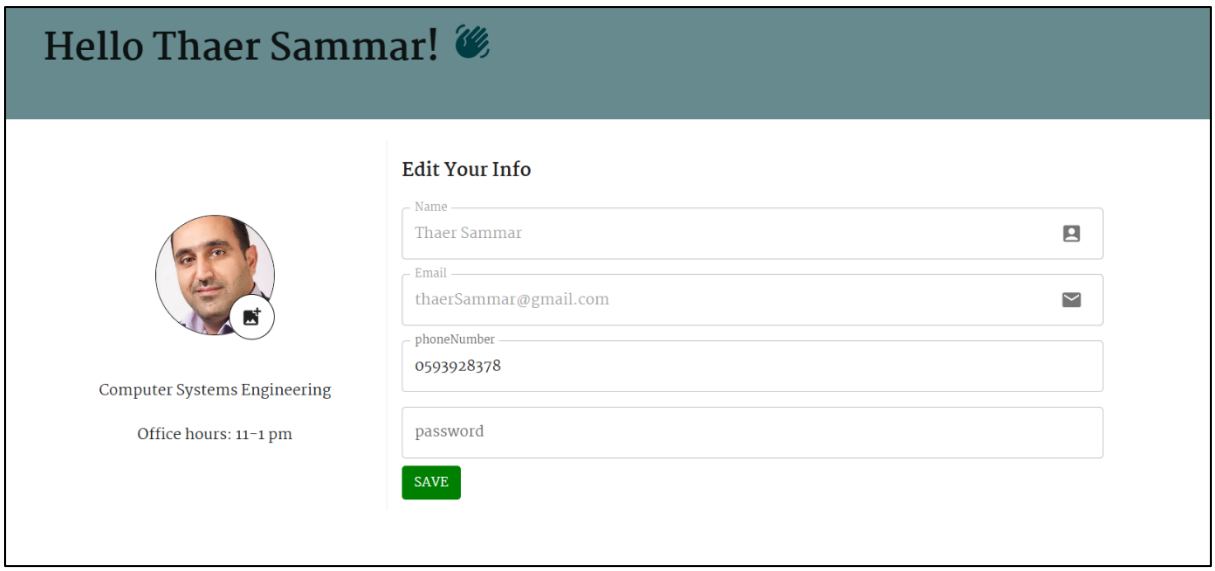

Figure 69: Supervisor profile

Figure 70: Supervisors' chat

Head of Department

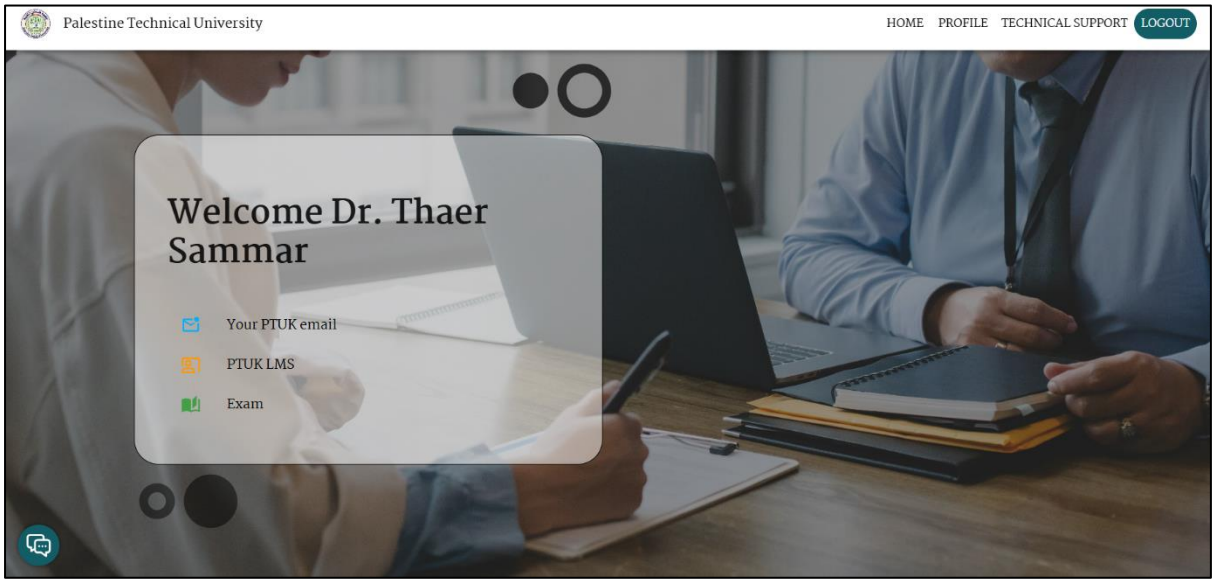

Figure 71: Head of department page header

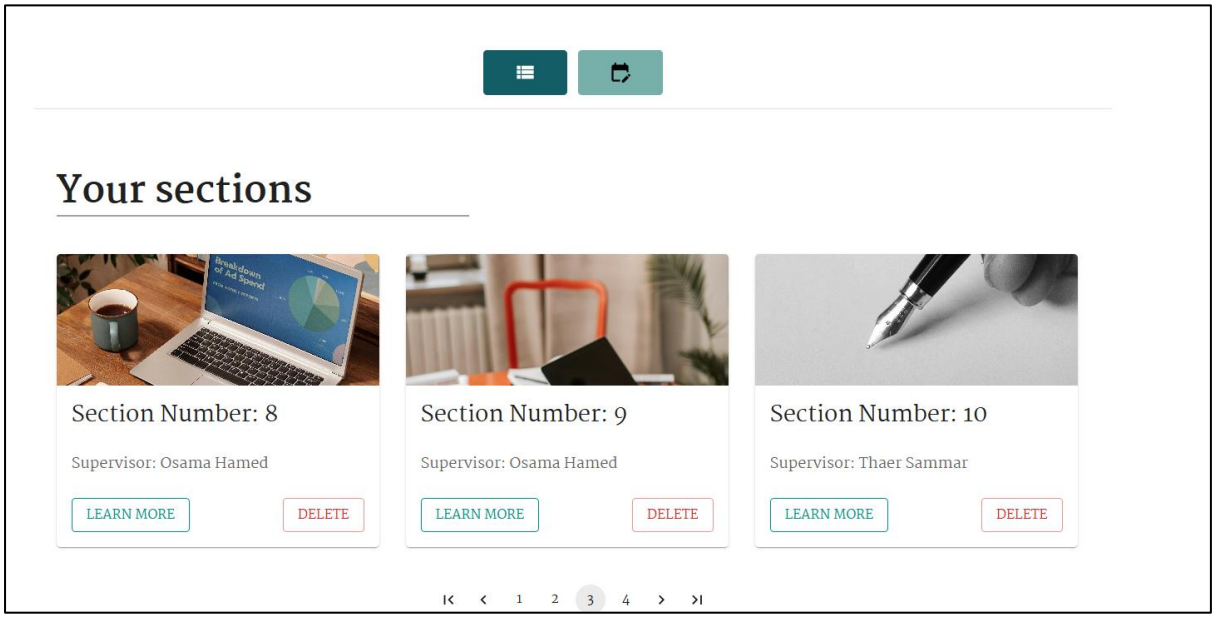

Figure 72: Head view sections tab

Figure 73: Add sections tab

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